



COURSE MATERIAL

for

PHYSICS

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Students are encouraged to complement this study guide with lectures, laboratory practicals, recommended textbooks, and other approved academic resources for a more comprehensive understanding of the course.

Published by:
Scholiquen Learning Platform

Website: <https://scholiquen.com>

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First Edition: 2026

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Study Guide

About This Study Guide

This Physics Study Guide is designed to assist students in understanding the fundamental concepts, principles, and applications of Physics. It provides a structured approach to learning the course contents and helps students prepare effectively for examinations and practical laboratory activities.

The guide contains the major topics, explanations, and study sessions covering important areas of Physics, including measurements, motion, forces, energy, fluids, and material properties. Each topic is arranged to provide a clear understanding of the concepts, equations, and practical applications required for academic success.

Students are encouraged to study each section carefully, practice calculations, review important definitions, and apply the knowledge gained through problem-solving exercises and laboratory experiments. Continuous revision and practical application are essential for developing a strong understanding of Physics.

Course Overview / Course Aim / Course Objectives

This course introduces students to the basic principles of Physics and develops their understanding of how physical laws govern natural phenomena. It provides knowledge of measurements, mechanics, forces, energy, fluids, and material behaviour.

The course aims to help students:

- Understand fundamental Physics concepts and principles.
- Develop skills in scientific measurement and laboratory practices.
- Apply mathematical methods to solve Physics problems.
- Explain motion, forces, energy transformations, and fluid behaviour.
- Understand the relationship between theoretical concepts and real-life applications.

Major topics covered include:

- Introduction to Physics and Laboratory Familiarization.
- Measurement of Mass, Weight, Length, and Time.
- Motion, Force, Circular Motion, and Friction.
- Scalar and Vector Quantities.
- Distance, Displacement, Speed, Velocity, and Acceleration.
- Motion Graphs and Uniform Acceleration Equations.
- Density and Relative Density.
- Upthrust, Archimedes Principle, Floatation, and Pressure.
- Work, Energy, Power, and Energy Conversion.
- Viscosity and Fluid Resistance.

Background to the Course

Physics is a fundamental science that explains the behaviour of matter, energy, and interactions in the universe. It provides the foundation for understanding technology, engineering, medicine, and many everyday processes.

This study material is designed to help students develop scientific thinking, analytical skills, and problem-solving abilities. Through the study of Physics, students will learn how to

observe physical phenomena, perform measurements, analyze data, and apply scientific principles to practical situations.

The course encourages students to connect classroom knowledge with laboratory experiments and real-world applications. Practical activities help students develop accuracy, observation skills, and confidence in handling scientific equipment.

Regular practice, revision, and self-assessment are important for mastering Physics concepts. Students should review equations, practice calculations, and attempt examination-style questions to strengthen their understanding.

Course Study Sessions

Session 1: Introduction to Physics and Laboratory Familiarization

Students will learn the meaning, scope, and importance of Physics. The session introduces laboratory equipment, safety procedures, measurement techniques, and scientific methods used in experiments.

Session 2: Physical Measurements

This session focuses on measuring mass, weight, length, and time. Students will understand standard units, measuring instruments, accuracy, and errors in measurements.

Session 3: Motion and Forces

Students will study motion in nature, different types of motion, force, circular motion, centripetal force, and centrifugal effects.

Session 4: Friction

This session explains friction, its causes, advantages, disadvantages, and methods of reducing friction in mechanical systems.

Session 5: Scalars, Vectors, and Motion Analysis

Students will examine scalar and vector quantities, distance, displacement, speed, velocity, acceleration, and graphical representation of motion.

Session 6: Velocity-Time Graph Calculations

Students will learn how to interpret velocity-time graphs and calculate distance, acceleration, and motion-related quantities.

Session 7: Density and Relative Density

This session introduces density concepts, calculations, relative density, and their applications in identifying materials.

Session 8: Fluids and Pressure

Students will study upthrust, Archimedes Principle, law of floatation, and pressure in fluids.

Session 9: Work, Energy, and Power

Students will explore work done, energy types, power calculations, energy conversion, and applications of energy in daily life.

Session 10: Viscosity

This session explains viscosity, fluid resistance, factors affecting viscosity, and practical applications of fluid flow.

Recommended Study Approach

Students should:

- Read each topic carefully.
- Understand definitions and principles.
- Practice Physics calculations regularly.
- Perform laboratory activities safely.
- Review equations frequently.
- Attempt self-assessment questions.
- Connect theories with practical examples.

Consistent study and continuous practice will improve understanding and performance in Physics examinations.

REFERENCE BOOKS

- New School Physics. By Prof. M.W Anyakoha
- New System Physics. By Dr. Charles
-

STUDY SESSION ONE - UNITS AND DIMENSIONS

1.0 INTRODUCTION

This session provides readers with a detailed explanation of the fundamental concepts of Physics, familiarization of Physics laboratory, physical quantities, units, and dimensions. It introduces the seven fundamental quantities, their units and symbols, together with examples of derived quantities and their respective units.

The session also explains the concept of dimensions of physical quantities, dimensional formulas, and the relationship between physical quantities and fundamental dimensions. In addition, it highlights the applications of dimensional analysis in checking equations, converting units, and solving physical problems.

Furthermore, the session introduces basic Physics laboratory equipment, their uses, measurement techniques, and safety practices required for effective laboratory investigations.

1.1 LEARNING OUTCOMES FOR THE STUDY SESSION 1

When you have studied this session, you should be able to:

- i. Identify the seven fundamental physical quantities and their SI units.
- ii. State the symbols and units of fundamental and derived quantities.
- iii. Explain the concept of dimensions and dimensional formulas.
- iv. Determine dimensional formulas of physical quantities.
- v. Apply dimensional analysis in Physics problems.
- vi. Identify common Physics laboratory equipment and their uses.
- vii. Explain basic laboratory safety procedures.

1.2 MEANING OF PHYSICS

Physics is one of the fundamental branches of science that seeks to explain how the universe works. It is the scientific study of matter, energy, and the interactions between them. Everything around us—from the movement of vehicles and the flow of electricity to the shining of the sun, the sound we hear, and the force of gravity—is governed by the principles of Physics. In simple terms, Physics helps us understand why things happen the way they do in nature.

Matter refers to anything that has mass and occupies space, while energy is the ability to do work or cause change. Physics studies how matter and energy interact in different forms. These forms of energy include motion (mechanical energy), heat, light, electricity, magnetism, sound, radiation, and gravitational energy. By understanding these interactions, scientists are able to explain natural phenomena and develop technologies that improve human life.

The scope of Physics is extremely broad. It examines objects of every size, from the smallest particles that make up atoms, known as subatomic particles, to the largest structures in the universe such as planets, stars, galaxies, and even the universe itself. Because of this wide scope, Physics provides the foundation for understanding many other scientific disciplines.

Physics can also be defined as the branch of natural science that studies matter, motion, force, energy, space, and time. It seeks to discover the laws that govern the physical world through careful observation, experimentation, measurement, and mathematical analysis. Rather than relying on assumptions, Physics explains natural events using logical reasoning and scientific evidence.

Historically, Physics is one of the oldest academic disciplines. In ancient times, the study of natural phenomena formed part of Philosophy, and subjects such as Mathematics, Chemistry, Biology, and Astronomy were closely connected. During the Scientific Revolution of the sixteenth and seventeenth centuries, Physics gradually developed into an independent scientific discipline with its own methods of investigation and experimentation. Today, it continues to serve as one of the foundations of modern science and technology.

The importance of Physics cannot be overemphasized. Almost every technological advancement in modern society is based on principles discovered through Physics. The study of electricity and magnetism has led to the invention of electric power systems, televisions, radios, computers, mobile phones, and many household appliances. The understanding of mechanics has made it possible to design vehicles, aircraft, bridges, and machines, while the study of thermodynamics has contributed significantly to engines, refrigeration, and power generation. Similarly, discoveries in nuclear physics have resulted in the development of nuclear energy, medical imaging equipment, and radiation therapy used in healthcare.

Physics also plays a vital role in advancing other fields of science and engineering. It provides the scientific principles upon which Chemistry, Biology, Medicine, Environmental Science, Engineering, Computer Science, and many other disciplines are built. As a result, a solid understanding of Physics is essential for students pursuing careers in science, technology, engineering, and related professions.

For ease of study, Physics is generally divided into two major branches:

i. Classical Physics

Classical Physics deals with physical phenomena that can be observed on a macroscopic scale. It includes the study of mechanics, heat, optics, waves and sound, electricity, and magnetism. These areas explain the behaviour of everyday objects and form the foundation of introductory Physics.

ii. Modern Physics

Modern Physics focuses on the behaviour of matter and energy at the atomic and subatomic levels. It includes topics such as quantum mechanics, atomic physics, nuclear physics, and relativity. These concepts help explain phenomena that cannot be adequately described by classical Physics.

Apart from these two major branches, several specialized fields of Physics have emerged to meet the needs of modern scientific research and technological development. These include:

- **Geophysics** – the application of Physics to the study of the Earth and its physical processes.
- **Astrophysics** – the application of Physics to the study of stars, planets, galaxies, and the universe.
- **Biophysics** – the use of Physics to understand biological systems and living organisms.
- **Nuclear Physics** – the study of atomic nuclei, radioactive materials, and nuclear reactions.

- **Engineering Physics** – the application of Physics principles to engineering design and technological innovation.

In conclusion, Physics is a dynamic and continually evolving science that seeks to explain the laws governing the physical universe. Its principles influence almost every aspect of daily life and continue to drive scientific discoveries and technological innovations that shape the modern world.

EVALUATION

1. What do you understand by the term "Physics"?
2. State the step involved in scientific method?

1.3 FUNDAMENTAL QUANTITIES AND UNIT

Measurement is a fundamental aspect of Physics because it allows physical quantities to be determined accurately and expressed in a standard form. Without measurement, it would be impossible to compare observations, perform experiments, or communicate scientific results effectively. Before any physical quantity can be measured, an appropriate unit of measurement must first be established to serve as a common standard.

Over the years, different systems of units have been developed and used in scientific studies. Some of the earlier systems include the **Foot–Pound–Second (FPS)** system, the **Centimetre–Gram–Second (CGS)** system, and the **Metre–Kilogram–Second (MKS)** system. As science and technology advanced, the need for a single international standard became necessary.

Today, the universally accepted system of measurement is the **International System of Units (SI)**, derived from the French term *Système International d'Unités (SI)*. The SI system provides a uniform set of units that is used worldwide in scientific research, engineering, education, and industry, ensuring consistency and accuracy in measurements.

In Physics, physical quantities are generally classified into **fundamental (base) quantities** and **derived quantities**. Fundamental quantities are the basic quantities that are defined independently, while derived quantities are obtained by combining two or more fundamental quantities through mathematical relationships.

1.4 FUNDAMENTAL QUANTITIES

Fundamental quantities, also known as **base quantities**, are the basic physical quantities from which all other physical quantities are derived. They are independent of one another, meaning that none of them can be expressed or defined in terms of any other physical quantity. These quantities serve as the foundation of the International System of Units (SI) and form the basis for measurement in Physics.

Every other physical quantity, known as a **derived quantity**, is obtained by combining one or more fundamental quantities through mathematical relationships. Therefore, a clear understanding of the fundamental quantities is essential for studying and applying the principles of Physics effectively.

1.5 FUNDAMENTAL UNITS: are the basic unit upon which other units depend. They are the units of the fundamental quantities.

The three most important basic quantities in Physics are length, mass and time.

Length may be defined as the extent of space or distance extended.

Mass is commonly defined as the quantity of matter or material in the body.

Time is defined as that in which events are distinguishable with reference to before or after. Examples of fundamental quantities and their units are shown below:

Table 2.0 Fundamental Quantities and Units

Quantity	Unit	Unit – abbreviation
Length	Metre	M
Time	Second	S
Mass	Kilogram	Kg
Electric current	Ampere	A
Temperature	Kelvin	K
Amount of substance	Mole	mol
Luminous intensity	Candela	Cd

1.6 DERIVED QUANTITIES AND UNITS

Derived quantities are physical quantities that are obtained by combining two or more fundamental quantities through mathematical relationships such as multiplication or division. Unlike fundamental quantities, derived quantities are not independent because they are expressed in terms of the fundamental quantities and their corresponding SI units.

The units assigned to these quantities are known as **derived units**. They are formed by combining the SI base units according to the mathematical relationship used to define each quantity. Derived quantities enable scientists to describe and measure a wide range of physical phenomena, including motion, force, energy, pressure, density, and power.

Some common examples of derived quantities and their corresponding SI units are presented in the table below.

Table 2.1 Derived quantities and units

Derived Quantity	Derivation	Derived unit
Area (A)	Length $\times$ breadth	m ²
Volume (V)	Length $\times$ breath $\times$ height	m ³
Density	$\frac{\text{mass}}{\text{volume}}$	Kg.m ⁻³
Velocity (V)	$\frac{\text{displacement}}{\text{time}}$	m.s ⁻¹
Acceleration (a)	$\frac{\text{change in velocity}}{\text{time}}$	m.s ⁻²
Force (F)	Mass $\times$ acceleration	Newton (N)

The unit of volume is obtained by multiplying three lengths $m \times m \times m = m^3$ pronounced 'CUBE METRE' or 'METRE CUBED'. Density is the ratio of mass and volume therefore the unit of density is kg/m³ or kgm⁻³ pronounced 'KILOGRAMME PER METRE CUBED.'

Difference between Fundamental and Derived Units

	Fundamental Units	Derived Units
1.	They are standard units of measurement	They are not standard units of measurement
2.	They are generally accepted all over the world	Not all are generally accepted all over the world
3.	They form the basis of measurement	They are not the basis of measurement
4.	They are accepted by international organisations	Though accepted internationally, they are formulated by individuals
5.	They are known as S.I. units, i.e. international system	They are known as units

Difference between Fundamental and Derived Quantities

	Fundamental Quantities	Derived Quantities
1.	They are generally accepted	They are just accepted
2.	They are based on international system	They are formulated from international system
3.	They can stand alone	They cannot stand alone
4.	They have direct calculations	Their calculations are derived
5.	They are basic units of measurement	They are not basic units of measurement

DIMENSIONS OF PHYSICAL QUANTITIES

The dimension of a physical quantity is the way it is related to the quantities of mass, length and time. The dimension of unit mass is M, for unit length, L and for unit time T. see the table below:

Table 2.2

Quantity	Unit	Dimension
Mass	Kilogramme, kg	M
Length	Metre, m	L
Time	Seconds, s	T

DIMENSION ANALYSIS OF SOME PHYSICAL QUANTITIES

1. Density: This is mass per unit volume

$$\begin{aligned} \text{The dimensional equation of density} &= \frac{\text{Dimension of Mass}}{\text{Dimension of Volume}} \\ &= \frac{\text{kg}}{\text{m}^3} = \frac{M}{L^3} = ML^{-3} \end{aligned}$$

2. Velocity: This is the rate of change of displacement with time.

$$\text{Velocity} = \frac{\text{Dimension of displacement (Length)}}{\text{Dimension of Time}} = \frac{m}{s} = \frac{L}{T} = LT^{-1}$$

3. Acceleration: This is the rate of change of velocity with time.

$$\text{Acceleration} = \frac{\text{Dimension of velocity}}{\text{time}} = \frac{ms^{-1}}{s} = \frac{LT^{-1}}{T} = LT^{-2}$$

4. Force: This is the product of mass and acceleration.

= Dimension of mass x Dimension of acceleration

$$= \text{kg} \times \text{ms}^{-2}$$

$$= \text{m} \times \text{LT}^{-2}$$

$$= \text{MLT}^{-2}$$

Below is a table of a few important physical quantities and their dimensions:

Table 2.3

Physical Quantity	Units	Dimensions
Velocity	ms ⁻¹	LT ⁻¹
Acceleration	ms ⁻²	LT ⁻²
Force	N(ma)	MLT ⁻²
Momentum	kgms ⁻¹	MLT ⁻¹
Density	kgm ⁻³	ML ⁻³
Pressure	Nm ⁻²	ML ⁻¹ T ⁻²

EVALUATION

1. State the dimension of the following quantities;
(a) Acceleration (b) pressure (c) density
2. From the following quantities given below, list out the derived and fundamental quantities in a tabular form: Velocity, mass, weight, length, volume, density, torque,

speed, acceleration, power, energy, temperature, heat capacity, electric current, relative density

Reading Assignment

New school physics by M.W.Anyakoha, Prof. Pg1-2

WEEKEND ASSIGNMENT

- Which of the units of the following physical quantities is not derived unit?
(a) Area (b) Thrust (c) Pressure (d) Mass
- Which of the following is a fundamental unit? (a) Kg m^{-3} (b) m^3 (c) Nm^{-2} (d) Kg
- Which of the following quantities has the same unit as energy?
(a) Power (b) Work (c) Force (d) Momentum
- Which of the following is a derived unit?
(a) Ampere (b) Kilogramme (c) Second (d) Ohm
- Which of the following is a derived unit?
(a) Tension (b) Impulse (c) Upthrust (d) Distance
- The international agreed system of unit (S.I.) for physical measurement are
(a) lb, ft, sec (b) g, m, sec (c) kg, m, sec (d) cm, g, sec
- Which of the units of the following physical quantities are derived?
I. Area; II. Thrust; III. Pressure; IV. Mass
(a) I, II, III and IV (b) I, II and III only (c) I, II and IV only (d) I and IV only

THEORY

- State the dimension of the following;
(a) stress (Force/Area) (b) Energy (force x perpendicular distance)
(c) Momentum (mass x velocity)
- Determine the dimension of the following physical quantities.
(a) Impulse (b) potential energy (c) pressure (d) young's modulus
- At what respective value of a, b, and c would the unit of impulse be dimensionally equivalent to $\text{M}^a \text{L}^b \text{T}^c$?
- The dimension of pressure is given as $\text{M}^x \text{L}^y \text{T}^z$, deduce the values of x, y, and z. (Hint: Pressure = Force/Area, the unit is Nm^{-2})

STUDY SESSION TWO - MEASUREMENT OF MASS, WEIGHT, LENGTH & TIME

INTRODUCTION

Measurement is one of the most important aspects of Physics because it allows us to describe physical quantities accurately and compare observations using standard units. Every scientific experiment, calculation, and engineering application depends on precise measurement. Without accurate measurements, it would be impossible to obtain reliable results or communicate scientific findings effectively.

In this session, you will learn the principles and techniques used in measuring four fundamental physical quantities: **mass, weight, length, and time**. You will become familiar with the instruments used for these measurements, understand the appropriate SI units, and develop the skills needed to take accurate readings while minimizing errors. A clear understanding of these measurement techniques forms the foundation for practical laboratory work and the study of more advanced concepts in Physics.

LEARNING OUTCOMES FOR THE STUDY SESSION

At the end of this session, you should be able to:

- i. Explain the importance of measurement in Physics and scientific investigations.
- ii. Distinguish clearly between mass and weight.
- iii. Identify the standard SI units for mass, weight, length, and time.
- iv. Recognize and correctly use common measuring instruments for each physical quantity.
- v. Measure mass, weight, length, and time accurately using appropriate instruments.
- vi. Record measurements correctly using the proper units and significant figures.
- vii. Identify common sources of measurement errors and apply methods to minimize them.
- viii. Interpret measurement results for solving basic Physics problems and laboratory experiments.

2.1 TECHNIQUES AND MEASUREMENT

Measurement is one of the most fundamental aspects of Physics and all other branches of science. Scientific facts, principles, and laws are only accepted when they can be observed, measured accurately, and expressed quantitatively. This makes measurement the foundation of scientific investigation, as it enables scientists to collect reliable data, compare results, and communicate findings using universally accepted standards. Every physical measurement consists of two essential components: a **numerical value**, which indicates the magnitude of the quantity being measured, and a **unit**, which provides the standard of comparison. For example, stating that a table tennis board is **20 metres long** combines the numerical value (**20**) with the unit (**metres**) to give a complete and meaningful measurement. Accurate measurement is therefore indispensable in Physics, ensuring precision in experiments, calculations, and practical applications.

2.2 MEASUREMENT OF LENGTH

THE METRE RULE: The metre rule is often used to measure distances of a few centimetres to some metres, for example, the dimensions of a table or room. When longer distances are involved, tape rule can be used. 0.1 cm or 1mm is the smallest graduation on a metre rule.

2.3 CALLIPERS:

Callipers are used to measure distances on solid objects where an ordinary metre rule cannot be applied directly. They are made of hinged steel jaws which are closed (in the case of external callipers) until they touch the desired part of the object being measured. The distance between the jaws is then measured on a graduated scale such as the metre rule.

2.4 THE VERNIER CALLIPERS

These can measure length more accurately than the metre rule. To measure small lengths, to the nearest 0.1mm, e.g the thickness of a metre rule, the internal and external diameters of a tube, or the diameter of a rod, we use the vernier callipers. The instrument has two sets of jaws and two scales, the main and the vernier scales

2.5 THE MICROMETER SCREW GAUGE

This instrument measures even smaller lengths (e.g diameter of a wire) than the vernier callipers. It has a higher reading accuracy and can read up to 0.01mm or 0.001cm. It can be used to measure the thickness of a piece of paper, the diameter of a small ball (e.g. pendulum bob).

2.6 MEASUREMENT OF TIME

The time interval between two events is the difference between the times when the event occurred when the time interval is of the order of minutes or hours, clocks and watches can be used. These are the instruments which indicate the time of the day. For shorter time intervals of the order of seconds, stop clocks or stop watches are used.

EVALUATION

1. Define (a) length (b) time.
2. List five instrument for measuring length.

2.7 MEASUREMENT OF MASS

The mass of a body is a measure of the quantity of matter it contains. Mass is usually measured by comparing it with standard masses, using a balance.

There are various types e.g. beam or chemical balance, lever balance, a dial spring, direct reading balance etc

2.8 WEIGHT: of a body is the force acting on the body due to the earth's gravitational pull. One instrument used for measuring weight is the spring balance. Weight is measured in Newtons.

Differences between Mass and Weight

	Mass	Weight
1.	Mass is the quantity of matter present in a body.	Weight occurs due to the force of gravity acting upon an object.
2.	Mass is constant.	Weight varies.
3.	Mass is a scalar quantity.	Weight is a vector quantity.
4.	The unit of mass is the kg.	The unit of weight is the Newton (N).
5.	Mass is measured by chemical or beam balance.	Weight is measured by spring balance.
6.	The principles of moment is applied to obtain the mass.	The instruments for measuring weight obeys Hook's law.

Relationship Between mass & weight

$$W=mg$$

Where, W = weight(N) m =mass(kg) g = acceleration due to gravity(m/s^2)

2.9 MEASUREMENT OF VOLUME

GRADUATED CYLINDER: A graduated cylinder can be used for measurement of volumes of liquids. It is accurate to the nearest $1cm^3$. It can also be used in measuring the volume of irregular shaped objects e.g stone, with the aid of the displacement or eureka can.

HOW TO READ A VERNIER CALLIPER

In reading a vernier calliper, the whole number (digit before the decimal point) and the first digit after the decimal point are read from the main scale while the second digit after the decimal

point is read from the vernier scale (sliding scale). This is the point or mark that coincides with that of the main scale.

See example below:

Example 3.1

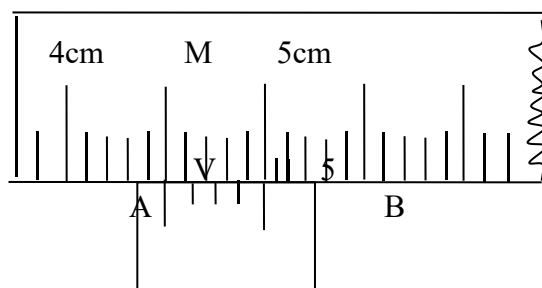


Fig. 3.3

Solution

What is the reading on the main scale = 4.30

Reading on the vernier scale = 0.07 i.e. point of coincidence = 0.07

∴ Total reading on the scale = 4.30

= 0.07

4.37cm

Note: The last digit before the end "A" of vernier inscribed on the main scale is 4.3 and the point of coincidence is 7 which is taken as 0.07. The rough end of the main scale indicates continuity, i.e the main scale is still extended.

Example 3.2 (How to read the micrometer scale)

What is the reading on the micrometer screw gauge shown below?

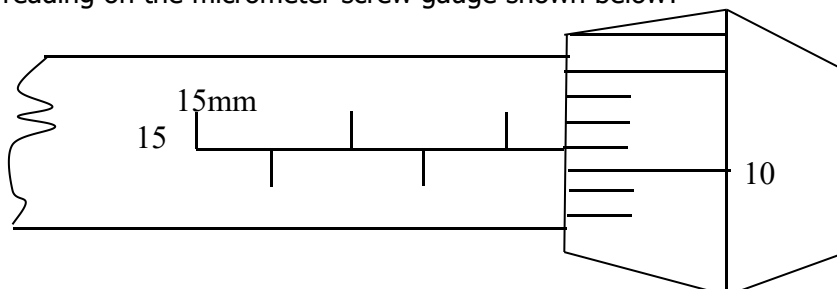


Fig. 3.4

Solution

Reading on the sleeve = 17.00

Reading on the thimble = 11 x 0.01

= 0.11

Reading on the scale = 17.1mm or 1.711cm

EVALUATION

1. Differentiate between mass & weight
2. The weight of an object of mass 5000g is (take $g = 10\text{m/s}^2$)

Reading Assignment

New school physics by M.W.Anyakoha, Prof. Pg3-11

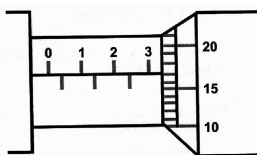
WEEKEND ASSIGNMENT

1. Which of the following instrument is suitable for taking the most accurate measurement of the internal diameter of a test-tube? (a) Metre rule (b) A pair of callipers (c) A micrometer screw gauge (d) A tape rule
2. Which of the following statements about mass and weight of a body is not correct? Its (a) mass is a scalar quantity (b) weight is a function of the gravitational pull on it (c) mass on earth and on the moon is the same (d) weight at the equator and at the poles is the same

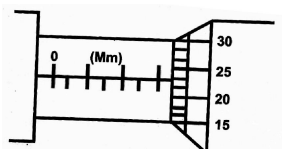
3. The diagram below represents a portion of a micrometer screw gauge. What is the reading? (a) 3.72mm (b) 3.50mm (c) 3.30mm (d) 3.25mm
4. The weight of a body is measured with
(a) spring balance (b) beam balance (c) chemical balance (d) lever balance
5. Which instrument is best for measuring small quantity of liquid?
(a) Burette (b) Pipette (c) Cylinder (d) Beaker
6. What is the dimension of force? (a) MLT^{-2} (b) ML^2T (c) ML^2T^{-2} (d) MLT^{-3}
7. The diagram below represents a portion of a vernier calliper. What is its reading?
(a) 4.36cm (b) 4.43cm (c) 5.53cm (d) 5.44cm
8. Which of the following instruments is the best for measuring the diameter of the constantan wire? (a) Callipers (b) Mere rule (c) Micrometer screw gauge (d) watt meter
9. Hydrometer is an instrument used for measuring
(a) relative humidity (b) dew point (c) relative density (d) attitude
10. What is the reading on the instrument? (a) 5.25mm (b) 10.16mm (C) 10.15mm (d) 5.10mm

THEORY

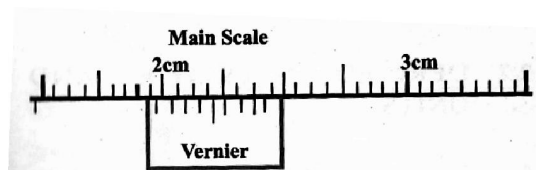
1. Mention the instrument that is suitable to measure the following quantities;
(a) The weight of a body (b) The internal diameter of a test tube
(c) The mass of a body (d) The dimension of a compound
(e) The length of a table (f) The diameter of a wire
2. The diagrams below represents the portion of a micro meter screw gauge. What are the readings?
(a)



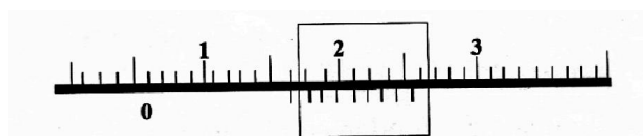
(b)



3. The length of a piece of glass block was measured by means of a vernier calliper as shown below. The length of the glass block is?



4. The diagram below represents a portion of a vernier calliper. What is the reading?



5. The internationally agreed system of units (SI) for physical measurements are _____, _____, and _____.
6. Mention three differences between mass and weight

STUDY SESSION THREE: MOTION IN NATURE

3.0 INTRODUCTION

Motion is one of the most fundamental concepts in Physics and is observed everywhere in nature. From the movement of planets around the Sun and the flow of rivers to the flight of birds and the movement of vehicles, every object in the universe is either in motion or capable of being set in motion by the action of forces. Understanding how and why objects move enables us to explain many natural phenomena and forms the basis for the study of mechanics.

In this session, you will examine the concept of **motion in nature**, the forces responsible for motion, and the different ways objects move under various conditions. You will also study **circular motion** and understand the roles of **centripetal** and **centrifugal forces** in maintaining circular paths. By the end of this session, you will be able to relate these concepts to everyday experiences and appreciate their applications in transportation, engineering, sports, astronomy, and other areas of science.

LEARNING OUTCOMES FOR THE STUDY SESSION

At the end of this session, you should be able to:

- i. Explain the concept of motion as it occurs in nature.
- ii. Distinguish between different types of motion.
- iii. Define force and explain its effects on the motion of objects.
- iv. Describe circular motion with practical examples.
- v. Explain the concepts of centripetal and centrifugal forces.
- vi. Differentiate between centripetal and centrifugal forces.
- vii. Apply the principles of motion and force to everyday situations and simple Physics problems.

3.1 FUNDAMENTALS OF MOTION

Many scientists have studied motion and its properties because of its importance to life. The Italian, Galileo Galilei, who lived from 1564 to 1642, did the first systematic study of motion. The science of the study of motion done by Galileo is known as kinematics. Isaac Newton was another scientist who did detailed work on the study of motion.

Motion involves a change of position of a body with time. It also involves how things move and what makes them to move. Kinematics is the description of how objects move without regard to forces causing their motion, and dynamics deals with why objects move as they do.

3.2 TYPES OF MOTION

There are four basic – types of motion. There are as follows.

1. **TRANSLATIONAL MOTION:-** When a body moves from a point A, along the line AB to another point B (see Fig. 4.1), we say that the body is translated from A to B, and the motion performed is known as translational motion. For example when an aeroplane flies from Abuja to Lagos or a car travels from Lagos to Enugu



Fig. 4.1 A horizontal line

2. **OSCILLATORY MOTION:** In this type of motion, a body moves to and fro, about a fixed point. Examples are the vibration of a plucked guitar string, the motion of a pendulum as it swings back and forth, the vibration of the molecules of a solid.
3. **RANDOM MOTION:** In this type of motion, the body moves in zigzag direction continuously so that they do not trace definite path. This type of motion is exhibited by molecules in gases. Other examples of random motion is the Brownian motion – an irregular motion of particles of various kinds suspended in water or smoke particles suspended in air e.t.c

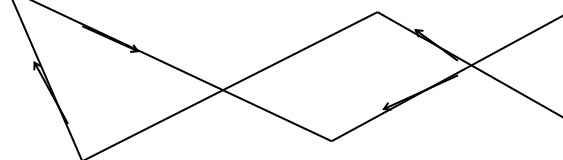


Fig 4.2 Random Motion

4. **ROTATIONAL MOTION:** This is the motion of a body which travels in a circle or ellipse and rotates about an axis. Examples are (i) the rotation of the earth about its axis (ii) the rotation of blades of an electric fan about its axis (iii) the rotation of a tap about a central axis.

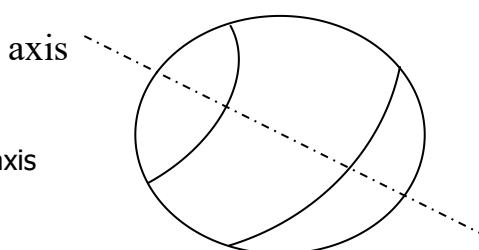


Fig. 4.3 The earth rotating on its axis

3.3 RELATIVE MOTION

If two bodies, A and B are moving on a straight line, the velocity of A relative to B is found by adding the Velocity of B reversed to the velocity of A. For instance, if a car traveling on a straight road at 100km/hr passes a bus going in the same direction at 60km/hr, the velocity of the car relative to the bus is $(-60+100) = 40\text{km/hr}$. If the car and the bus are traveling in opposite direction with the same velocities of 100km/r and 60km/hr respectively, the velocity of the car relative to the bus is $(-(-60) + 100) = (60 + 100) = 160 \text{ km/hr}$.

NB: When the velocities are not in the same straight line, the parallelograms law should be used to add this since velocities are vectors, and their magnitudes and direction must be taken into consideration.

3.4 CAUSES OF MOTION

We have been describing the motion of a body without regard to what causes the motion. A block of wood resting on a table will remain at rest until it is pushed or pulled by an agent. Such an agent that change or tends to change the state of rest or uniform motion in a straight line of a body is called **force**.

3.5 TYPES OF FORCE

There are two main types of forces, contact force and force field.

1. **CONTACT FORCE:** This may be regarded as a force which exists between surfaces in contact. It includes pushing and pulling forces, frictional forces, reaction and tension forces in strings and wires.
2. **FORCE FIELDS:** These are forces whose sources do not require contact with the body to which they are applied. Examples are gravitational force, electrostatic and magnetic forces

EVALUATION

1. Explain the types of motion.
2. Differentiate between contact & field force.

3.6 SIMPLE IDEA OF CIRCULAR MOTION

An object moving with a constant speed along a circular path is said to have a uniform circular motion. Examples are the moon circling the earth, the planets moving round the sun, earth moving round the sun, stone tied to a string which is whirled in a horizontal vertical circle.

3.7 Circular motion has three characteristics:

- (1) constant speed (2) changing or variable velocity (3) centripetal acceleration.

The acceleration that is directed towards the centre of the circular path is known as centripetal acceleration. Its magnitude a is given by $a = \frac{v^2}{r}$

Where V is the uniform speed and r is the radius of the circular path.

Centripetal force F_T is defined as that inward force required to keep an object moving with a constant speed in a circular path

The centripetal force is given by $F_T = \frac{mv^2}{r}$ where m is the mass of the object moving with a uniform velocity v in a circular path or radius r .

Centrifugal force: The centrifugal force is the reaction force that tends to move a body away from the centre. In other words, it acts in opposite direction to the centripetal force $\left(\frac{-mv^2}{r}\right)$

Centrifuge: A centrifuge is a device used to separate particles in suspension from the liquid in which they are contained.

EVALUATION

1. Differentiate between centripetal & centrifugal force.
2. A stone tied to a string is made to revolve in a horizontal circle of radius 4m with an angular speed of 2 rad/s. With what is tangential velocity will the stone move off the circle if the string cuts?

READING ASSIGNMENT

www.google.com(click on google search, type "circular motion",click on search) & - New school physics by M.W.Anyakoha,Phd. Pg 12-27

WEEKEND ASSIGNMENT

1. Which of the following types of motion does a body undergo when moving in a haphazard manner? (a) Random motion (b) Translatory motion (c) Rotational motion (d) Vibratory motion
2. What type of motion does the skin of a talking drum perform when it is struck with a drum stick? (a) Rotational (b) Translational (c) Random (d) Vibratory
3. The motion of the prongs of sounding tuning fork is (a) rotational (b) vibratory (c) vibratory and rotational (d) random
4. A body moves with a constant speed but has an acceleration. This is possible if it (a) moves in a straight line (b) moves in a circle (c) is oscillating (d) is in equilibrium (e) has a varying
5. A body moves along a circular path with uniform angular speed of 0.6rad s^{-1} and at a constant speed of 3.0ms^{-1} . Calculate the acceleration of the body towards the centre of the circle.
(a) 25.0ms^{-2} (b) 5.4ms^{-2} (c) 5.0ms^{-2} (d) 1.8ms^{-2}
6. The angular speed of an object describing a circle of radius 4m with a linear constant speed of 10ms^{-1} is (a) 40rad s^{-1} (b) 14rad s^{-1} (c) 2.5rad s^{-1} (d) 0.40rad^{-1}
7. A body moving at a constant speed accelerates when it is in (a) rectilinear motion (b) translational motion (c) circular motion (d) vibrational motion
8. The study of motion without involving the force which causes the motion is called (a) kinematics (b) inertia (c) electromagnetic (d) dynamics

9. The magnitude of the force required to make an object of mass M move with speed V in a circular path of radius R is given by the expression
 (a) $\frac{mv^2}{r}$ (b) $\left(\frac{mv^2}{r}\right)^2$ (c) $\frac{mv^2}{r}$ (d) $\frac{mv}{2}$
10. The following are types of motion except (A) random motion (b) rotational motion (c) nuclear motion (d) oscillatory motion.
11. The motion of the prongs of a sounding tuning fork is (a) random (b) translational (c) rotational (d) vibratory
12. A body moving in a circle at constant speed has
 i. a velocity tangential to the circle
 ii. a constant kinetic energy
 iii. an acceleration directed towards the circumference of the circle. Which of the statement above are correct
 (a) i & ii only (b) ii & iii only (c) i & iii only (d) i, ii & iii
13. A loaded test – tube which floats upright in water is carefully and slightly depressed and then released. which of the following best describes the subsequent motion of the test tube (a) circular (b) rotational (c) random (d) oscillatory.
14. Which of the following correctly gives the relationship between linear speed v & angular velocity w of a body moving uniformly (a) $v=w r$ (b) $v=w^2r$ (c) $v=wr^2$ (d) $v=w/r$

THEORY

1. Mention and describe two practical situations where centripetal force must be taken into account.
2. A body weighing 100N with a speed of 5ms^{-1} in a horizontal circular path of radius 5m. Calculate the magnitude of the centripetal force acting on the body ($g= 10\text{ms}^{-2}$). (WAEC, 1999)
3. A piece of stone attached to one end of spring is whirled round in a horizontal circle of radius 7m. When the constant speed of the stone is 40ms^{-1} , calculate the centripetal acceleration.
4. A keke (tricycle) moves around Mary Slessor roundabout of radius 50m, at a constant speed of 20ms^{-1} , find (a) Centripetal acceleration (b) Centripetal force
5. A particle of mass 100kg is fixed to the tip of a fan blade which rotates with angular velocity of 100rads^{-1} . If the radius of the blade is 0.2m, find the centripetal force.
6. A body of mass 5kg moving in a circular path with a velocity of 5m/s for 10 complete revolution within 4s. If the radius of the circular path is 30m. Find (a) the centripetal force (b) the centripetal acceleration (c) angle subtended in radian (d) angular velocity

STUDY SESSION FOUR: FRICTION

4.0 INTRODUCTION

Friction is a force that opposes the relative motion or the tendency of motion between two surfaces in contact. It is an everyday phenomenon that affects nearly every activity, from walking and writing to driving vehicles and operating machines. Although friction is often viewed as a force that slows down motion, it is also essential because it provides the grip needed for movement and prevents objects from slipping.

In this session, you will study the concept of friction, the conditions under which it occurs, and the different types of friction encountered in everyday life. You will also examine the factors that affect friction, its advantages and disadvantages, and the practical methods used to increase or reduce it in various scientific, industrial, and engineering applications. Understanding friction is important because it helps explain the behavior of moving objects and improves the design and efficient operation of machines and equipment.

LEARNING OUTCOMES FOR THE STUDY SESSION

At the end of this session, you should be able to:

- i. Define friction and explain how it arises between surfaces.
- ii. Identify the different types of friction.
- iii. State the factors affecting the magnitude of frictional force.
- iv. Explain the advantages and disadvantages of friction.
- v. Describe practical methods of increasing and reducing friction.
- vi. Apply the principles of friction to everyday activities and engineering applications.
- vii. Solve simple problems involving frictional forces.

DEFINITION OF FRICTION

Friction (F_r) is defined as a force which acts at the surface of separation between two objects or two bodies in contact and tend to oppose the motion of one over the other. It is simply force of opposition. We have two types of friction:

- (a) Static friction, F_s
- (b) Dynamic friction, F_d . F_s is greater than F_d

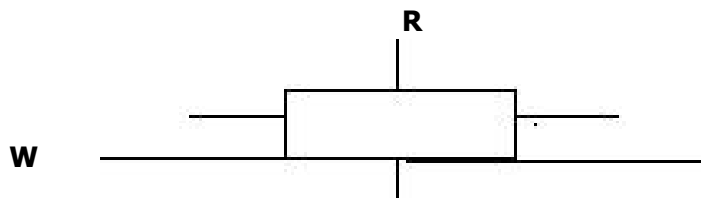
LAWS OF SOLID FRICTION

1. Friction opposes the relative motion of two surfaces in contact.
2. It is independent of the area of the surface of contact.
3. It depends on the nature of the surface.
4. It is proportional to normal reaction (R).
5. It is independent of relative velocity between the surfaces

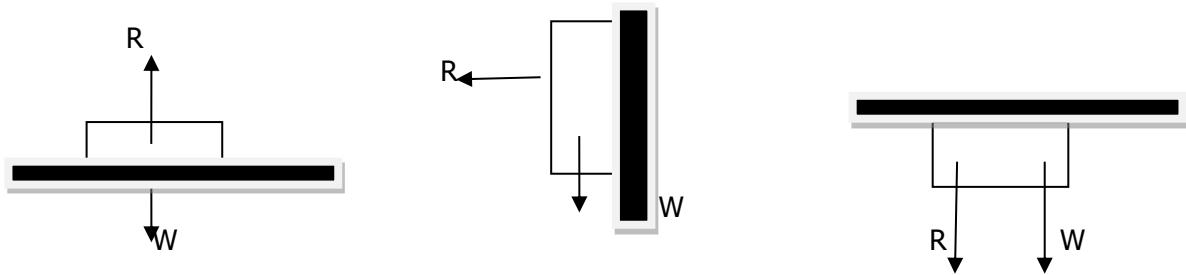
$$F_r \propto R$$

$$F_r = \mu R \dots\dots\dots 1.$$

where F_r -frictional force μ -coefficient of friction & R -normal reaction



The weight (W) of an object is acting vertically downward.. the normal reaction (R) is always acting perpendicular to the plane.. the normal reaction is equal to the weight.



$$W = mg$$

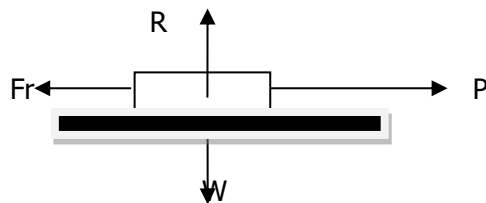
At equilibrium, $R = mg$, this implies that,

$$R = W \quad [g \text{ is acceleration due to gravity} = 10\text{m/s}^2]$$

$$R = mg$$

$$F = \mu mg$$

$$\mathbf{Fr = \mu mg} \dots\dots\dots 2$$



Case one: if the force P is applied, and the object is stationary.

$$P - Fr = ma$$

Since no motion $a = 0$

$$P - Fr = 0$$

$$\mathbf{P = Fr} \dots\dots\dots 3$$

Case two: when the force P is applied and the body moves.

$$P - Fr = ma$$

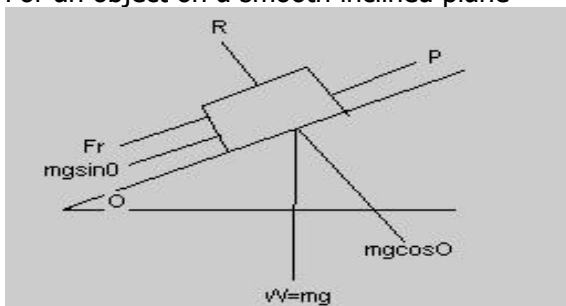
$$P = Fr + ma$$

$$\text{But } Fr = \mu mg$$

$$P = \mu mg + ma$$

$$\mathbf{P = m [\mu g + a]} \dots\dots\dots 4$$

For an object on a smooth inclined plane



Case one: if the body moves upward, $a > 0$

$$P - mgsin\theta = ma$$

$$P = mg \sin \theta + ma \dots\dots\dots 5$$

Case two: if the body is stationary $a = 0$

$$P - mg \sin \theta = ma$$

$$P - mg \sin \theta = 0$$

$$P = mg \sin \theta \dots\dots\dots 6$$

Case three: if the body slides down the plane, $a > 0$

$$mg \sin \theta - P = ma$$

$$P = ma + mg \sin \theta \dots\dots\dots 7$$

For a body on a rough inclined plane.

$$P - mg \sin \theta - Fr = ma$$

$$\text{But } Fr = \mu mg$$

$$P - mg \sin \theta - \mu mg = ma \dots\dots\dots 8$$

$$\text{Also, } R = mg \cos \theta$$

$$P - mg \sin \theta - \mu mg \cos \theta = ma \dots\dots\dots 9$$

If the body moves upward the incline plane

$$\mu = \tan \theta \dots\dots\dots 10$$

EVALUATION

1. Differentiate between static and dynamic friction.
2. State the laws governing solid friction.

ADVANTAGES OF FRICTION

- (1) It makes walking and running possible.
- (2) It enables gripping of belt in machines possible.
- (3) It enables nails to stay in the wall when driven.
- (4) It stops tires from slipping.
- (5) Enable cars to stop when breaks are applied.
- (6) Enables human to use mouse in surfing web.

DISADVANTAGES OF FRICTION

- (1) Causes wear and tear.
- (2) Causes the efficiency of the machines.
- (3) Causes a lot of energy to be consumed by the machine.
- (4) Causes loss of resources.

REDUCING FRICTION

1. Lubricating surfaces with grease, oil etc.
2. Using ball or roller on wheels.
3. Smoothing or polishing the surface.
4. By streamlining.

EVALUATION

1. State three (3) advantages & two (2) disadvantages of friction.
2. State three (3) ways of reducing friction.

READING ASSIGNMENT

www.google.com (click on google search, type "what is friction", click on search) New school physics by M.W .Anyakoha,Phd. Pg 19 – 28.

WEEKEND ASSIGNMENT

1. Friction depends on the area of surface in contact (a) true (b) false (c) true & false (d) none of the above

2. A metal block of mass 8kg lies on a rough horizontal platform. If the horizontal resistive force is 10N, find the coefficient of static friction ($g=10\text{m/s}^2$) (a) 0.25 (b) 0.125 (c) 0.8 (d) 0.124
3. Which of the statement is correct (a) static friction is less than dynamic friction (b) static friction equals dynamic friction (c) static friction is greater than dynamic friction (d) none of the above
4. A metal block of mass 0.5kg lies on a rough horizontal plane, what is the normal reaction ($g=10\text{m/s}^2$) (a) 50N (b) 0.05N (c) 500N (d) 5N
5. If the angle between the incline length and the horizontal platform of an incline plane is 300 calculate the coefficient of friction (a) 0.542 (b) 0.577 (c) 0.467 (d) 0.866

THEORY

1. Define friction and state the laws governing solid friction.
2. A body of weight 6N rest on a plane inclined at an angle of 300 to the horizontal (a) what force keeps it sliding down the plane? (b) what is the coefficient of friction?
3. State **two**
 - (i) Laws of friction
 - (ii) Advantages of friction.
 - (iii) Methods of reducing friction (WAEC,2006)
4. A force, 10N drags a mass 10kg on a horizontal table with an acceleration of 0.2ms^{-2} . If the acceleration due to gravity is 10ms^{-2} . Calculate the coefficient of friction between the moving mass and the table. (UME,1998)

STUDY SESSION FIVE: VECTOR & SCALAR QUANTITY, DISTANCE/DISPLACEMENT, SPEED/VELOCITY, ACCELERATION, DISTANCE/DISPLACEMENT –TIME GRAPH, SPEED/VELOCITY–TIME GRAPH

1.0 INTRODUCTION

This session introduces some of the most important concepts in mechanics that are used to describe and analyze motion. It begins by distinguishing between **scalar and vector quantities**, helping learners understand the role of magnitude and direction in physical measurements. The session also explains the concepts of **distance and displacement**, as well as **speed and velocity**, highlighting their similarities and differences in describing motion.

In addition, the session examines **acceleration** as the rate at which velocity changes with time and demonstrates how the motion of objects can be represented graphically using **distance–time** and **speed/velocity–time graphs**. These graphs provide a visual interpretation of motion and make it easier to determine important quantities such as speed, velocity, acceleration, and distance traveled. A sound understanding of these concepts forms the foundation for solving a wide range of problems in mechanics and for understanding more advanced topics in Physics.

LEARNING OUTCOMES FOR THE STUDY SESSION

When you have studied this session, you should be able to:

- i. Differentiate between scalar and vector quantities.
- ii. Distinguish between distance and displacement.
- iii. Explain the concepts of speed and velocity.
- iv. Define acceleration and describe its significance in motion.
- v. Interpret and analyze distance–time graphs.
- vi. Interpret and analyze speed/velocity–time graphs.
- vii. Apply these concepts to solve simple problems involving motion.

5.0 SCALAR & VECTOR QUANTITY

5.1 A scalar quantity is defined as a quantity that has magnitude only but no direction. Typical examples of scalar quantities are time, distance, speed, temperature, volume, work, power, electric potential etc. A scalar quantity or parameter has no directional component, only magnitude. For example, the units for time (minutes, days, hours, etc.) represent an amount of time only and tell nothing of direction. Additional examples of scalar quantities are density, mass, and energy.

5.2 A vector quantity is defined as a quantity that has both magnitude and direction. Typical examples of vector quantities are velocity, displacement, acceleration, force, momentum, moment, electric field intensity etc

5.3 POSITION

Position is referred to as the point in which an object can be located or the place object is found. The position of an object on a plane can be given by its co-ordinates, i.e., the signed distances of the point from two perpendicular axes, OX and OY

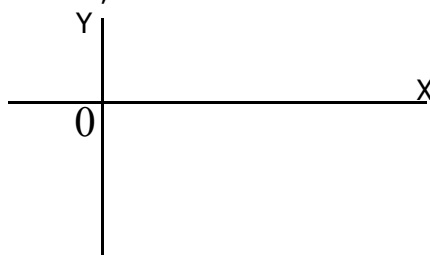


Fig. 6.0 Cartesian co-ordinates

The x – co-ordinates is called abscissa while the y – co-ordinate is called ordinate. The x co-ordinate is written first, before the y – co-ordinates, i.e. (X,Y)

5.4 DISTANCE AND DISPLACEMENT

Distance: This is the gap between any two positions in space. It is denoted by S and measured in metre(m) it is a scalar quantity and is calculated as the product of average speed and time. Thus, distance = average speed $\times$ time.

Displacement: This is the distance covered in a specific direction. it is a vector quantity measured in metre(m). The direction of motion of bodies can be found by using the compass. Displacement = average velocity $\times$ time. It is denoted by X

The Use of Bearing to Indicate Direction and Displacement

The bearing of an object from the origin is the angle which it makes with the north pole in the clockwise sense. It is specified in two ways:

- The use of cardinal points: N – North, S – South, W – West, and E – East
- The use of three digit notation. Students should note that bearing which are located by cardinal points are with respect or reference to the North and South.

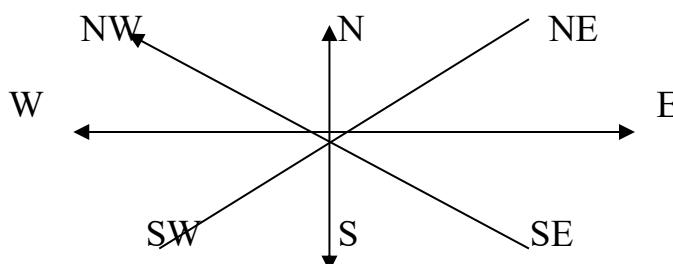


Fig. 6.1 cardinal points and their directions

5.5

SPEED AND VELOCITY

Speed: Speed is defined as the rate of change of distance moved in an unspecified direction or the rate of change of distance per unit time in an unspecified direction. It is measured in metre per second (m/s). It is a scalar quantity.

The mathematical expression of speed is

$$\text{Speed} = \frac{\text{Distance moved}}{\text{Time}}$$

Average Speed: Average speed is defined as the ratio of the total distance travelled to the total time taken. It is a scalar quantity and measured in m/s or ms^{-1}

This, average speed = $\text{Average Speed} = \frac{\text{Total Distance travelled}}{\text{total time taken}}$

When a body covers equal distance in equal time intervals, no matter how small the time interval may be, it is said to be a **uniform speed or constant speed**.

Velocity: Velocity is defined as the rate of change of distance moved in a specific direction or the rate of change of displacement. Velocity is a vector quantity. For instance, it would be easy and correct to say that a car travelling at a steady speed of 50km/h in a direction of N40°E has a velocity of 50km/h, N40°E.

$$\text{velocity} = \frac{\text{Displacement}}{\text{Time}}$$

Uniform velocity

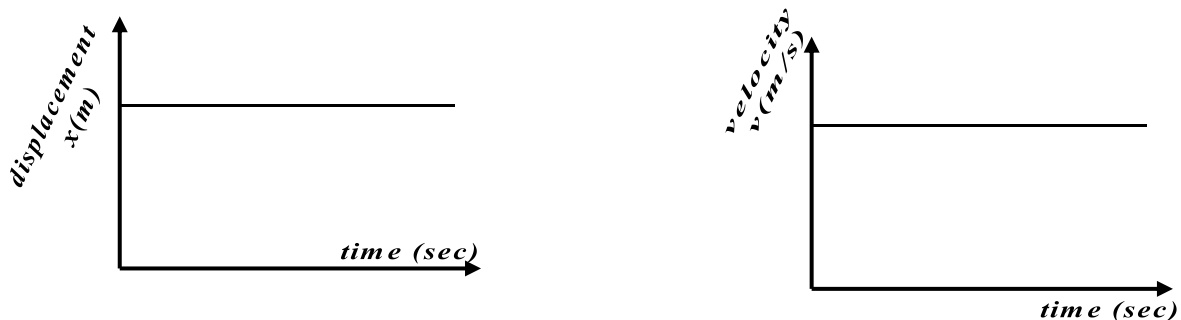


Fig 5.6 Uniform Velocity

Uniform (constant) velocity: An object is said to undergo (constant) velocity, if the rate of change of displacement is constant, no matter how small the interval may be.

Example 1:

A train moves with a speed of 54km/h for one quarter minute. Find the distance travelled by the train.

Solution:

$$\begin{aligned} \text{Speed} &= 54\text{km/h} = 15\text{m/s} \\ \text{Time} &= \frac{1}{4} \text{ min} = \frac{1}{4} \times 60 = 15\text{s} \\ \text{Distance} &= \text{speed (m/s)} \times \text{time (s)} \\ &= 15(\text{m/s}) \times 15(\text{s}) \\ &= 225\text{m} \end{aligned}$$

5.7 ACCELERATION & RETARDATION

Acceleration is defined as the increasing rate of change of velocity. It is measured in m/s^2 .

$$\text{Acceleration (a)} = \frac{\text{Increasing Velocity change}}{\text{Time taken}}$$

When the velocity of a moving body increases by equal amount in equal intervals of time, no matter how small the time intervals may be, it is said to move with **uniform acceleration**.

Retardation is defined as the decreasing rate of change of velocity. It is measured in m/s^2 . It is also known as deceleration or negative acceleration

$$\text{Retardation (ar)} = \frac{\text{Decreasing Velocity Change}}{\text{Time Taken}}$$

5.8 EQUATION OF UNIFORMLY ACCELERATED MOTION

$S = \frac{(v+u)}{2} t$	7
$v = u + at$	8
$v^2 = u^2 + 2aS$	9
$S = ut + \frac{1}{2} at^2$	10

Equations (7) to (10) are called equations of uniformly accelerated motion and could be used to solve problems associated with uniformly accelerated motion

where u - initial velocity(m/s), v – final velocity (m/s), a – acceleration (m/s^2), s – distance covered and t – time (m).

Example 2

A car moves from rest with an acceleration of $0.2m/s^2$. Find its velocity when it has moved a distance of 50m.

Solution:

$a = 0.2m/s^2$, $S = 50m$, $u = 0m/s$, $v = ?$

$v^2 = u^2 + 2aS$

$v^2 = 0^2 + (2 \times 0.2 \times 50) = 20$

$v = \sqrt{20} \text{ m/s}$

EVALUATION

1. State the differences & similarity between speed & velocity.
2. A car has a uniform velocity of 108km/hr. How far does it travel in $\frac{1}{2}$ minute?

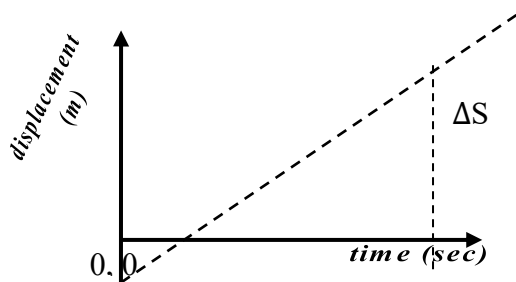
GRAPHS

The motion of an object is best represented or described with graphs. These graphs are

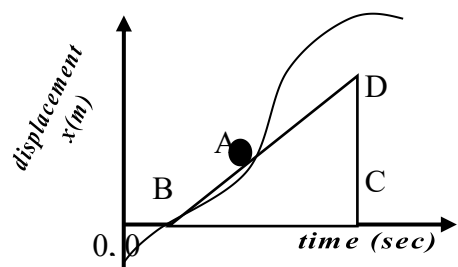
- i) Distance- time
- ii) Displacement – time
- iii) Velocity – time

Distance – time

In a distance-time graph, its slope or gradient gives the speed.



(i) Uniform speed



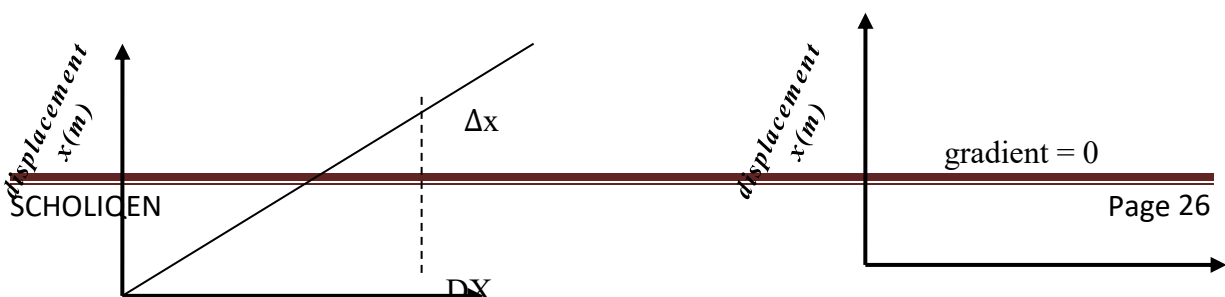
(ii) Non-uniform speed

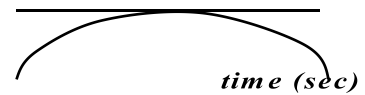
Fig. 6: Distance-time graph

$$\text{Gradient/slope} = \text{speed} = \frac{\text{distance}}{\text{time}} = \frac{\Delta s}{t}$$

Displacement – time graph

A displacement-time graph could be linear or curved. For a linear graph, the gradient gives the velocity.





a) Non-uniform velocity

time (sec)

Fig. 6.4 Displacement-time graph

$$\text{Gradients/slope} = \text{velocity (v)} = \frac{\text{displacement}}{\text{time}} = \frac{\Delta x}{t}$$

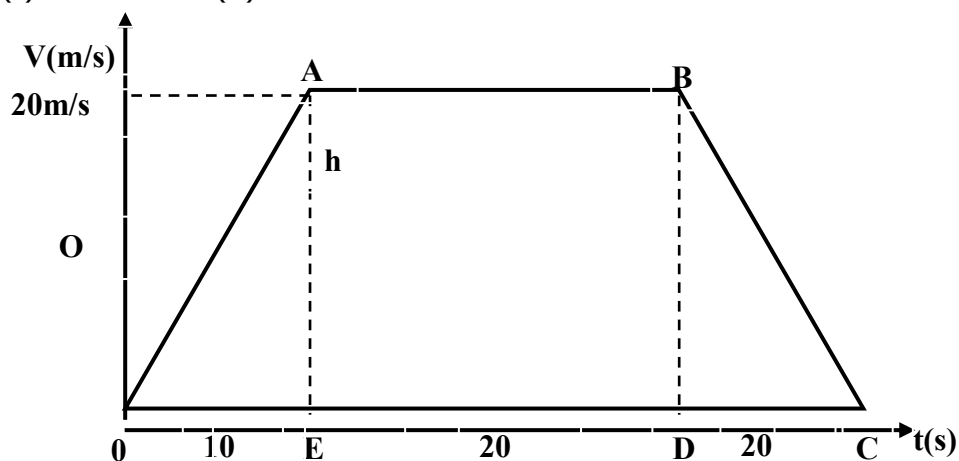
Velocity – time graph

The velocity-time graph is more useful than any of the two graphs described above because it gives more useful information concerning the motion of objects. The following information can be obtained from the graphs (i) acceleration (ii) retardation (iii) distance (iv) average speed.

The motion of objects can form shapes such as square, triangle, trapezium, rectangle or a combination of two or more shapes. Thus, the sum of the areas of the shapes formed corresponds to the distance moved, covered or travelled by the objects.

Example 3

A motor car accelerates for 10secs to attain a velocity of 20m/s. It continues with uniform velocity for a further 20 seconds and then decelerates so that it stops in 20 seconds. Calculate (i) Acceleration (ii) Deceleration (iii) The distance travelled.



$$\begin{aligned} \text{i)} \quad V &= u + at, u = 0 & \text{or} \quad a &= \frac{AE}{OE} = \frac{20}{10} = 2\text{m/s}^2 \\ 20 &= 10a \\ a &= \frac{20}{10} = 2\text{m/s}^2 \end{aligned}$$

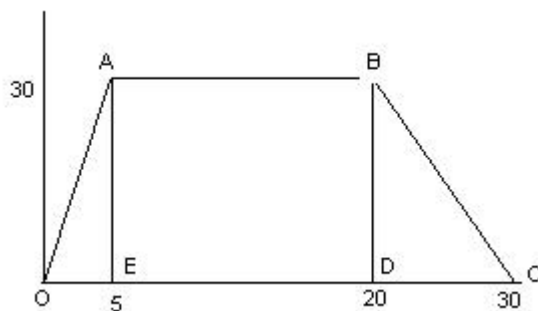
$$\text{ii)} \quad \text{Deceleration} = \frac{\Delta V}{t} = \frac{0 - 20}{20} = -1\text{m/s}^2 \quad \text{i.e.} \quad \frac{BD}{DC} = \frac{20}{20} = -1\text{m/s}^2$$

$$\begin{aligned} \text{iii)} \quad \text{Using area of trapezium} \\ \frac{1}{2} \times (AB + OC) h &= \frac{1}{2} \times (20 + 50) 20 \\ = \frac{1}{2} \times (70) \times 20 &= 700\text{m} \end{aligned}$$

Example 4

A car starts from rest and accelerates uniformly until it reaches a velocity of 30m/s after 5 seconds. It travels with uniform velocity for 15 seconds and is then brought to rest in 10s with a uniform retardation. Determine (a) the acceleration of the car (b) The retardation (c) The distance covered after 5s (d) The total distance covered (use both graphical and analytical method).

The velocity – time diagram for the journey is shown above, from this diagram



- a. the acceleration = slope of OA
 $= AE / EO$
 $= (30-0) / (5-0) = 30/5$
 $= 6 \text{ m/s}^2$
- b. the retardation = slope of BC = CB / CD
 $= (0-30) / (30-20) = -30/10$
 $= -3 \text{ m/s}^2$ (the negative sign indicates that the body is retarding)
- c. Distance traveled after 5s = area of AEO
 $= \frac{1}{2} \times b \times h$
 $= \frac{1}{2} \times 5 \times 30$
 $= 75 \text{ m}$
- d. Total distance covered = area of the trapezium OABC
 $= \frac{1}{2} (AB + OC) AE$
 $= \frac{1}{2} (15 + 30) 30$
 $= 675 \text{ m}$

Using equations of motion.

- a) $U = 0, V = 30, t = 5$
 $V = u + at$
 $a = (v-u)/t = (30-0)/5$
 $a = 30/5 = 6 \text{ m/s}^2$
- b) a or in
 $a = (v-u)/t = (0-30)/10$
 $a = -3 \text{ m/s}^2$

$$(c) S = \left(\frac{u+v}{2} \right) t$$

$$= \left(\frac{30+0}{2} \right) 5$$

$$= 75 \text{ m}$$

(d) To determine the total distance travelled, we need to find the various distance for the three stages of the journey and then add them.

for the 1st part $S = 75 \text{ m}$ from (c)

for the 2nd stage where it moves with uniform velocity.

$$S = vt$$

$$= 30 \times 15$$

$$= 450 \text{ m}$$

for the last stage $S = \frac{1}{2} (u + v) t$
 $= \frac{1}{2} (30 + 0) 10$
 $= 150 \text{ m}$

$$\text{Total distance} = 75 + 450 + 150 = 675 \text{ m}$$

EVALUATION

- A train slows from 108km/hr with uniform retardation of 5 m/s^2 . How long will it take to reach 18km/hr and what is the distance covered?
- Why is velocity – time more useful than displacement time graph?

READING ASSIGNMENT

WEEKEND ASSIGNMENT

1. A body which is uniformly retarded comes to rest in 10s after travelling a distance of 20m. Calculate its initial velocity (a) 0.5 ms^{-1} (b) 2.0 ms^{-1} (c) 4.0 ms^{-1} (d) 20.0 ms^{-1} (e) 200.0 ms^{-1}
2. The distance travelled by a particle starting from rest is plotted against the square of the time elapsed from the commencement of the motion. The resulting graph is linear. The slope of the graph is a measure of (a) initial displacement (b) initial velocity (c) acceleration (d) speed
3. Which is the correct formula for a body accelerating uniformly?
(a) $a = \frac{v^2 - u^2}{2s}$ (b) $v^2 = u^2 + 2as$ (c) $s = \frac{1}{2}ut + at^2$
(d) $v^2 - u^2 = 2as$ (e) $v = u + at$
4. The slope of a displacement-time graph is equal to
(a) acceleration (b) uniform velocity (c) uniform speed (d) instantaneous speed
5. A body moving with uniform acceleration has two points (5, 15) and (20, 60) on the velocity-time graph of its motion. Calculate (a) 0.25 ms^{-2} (b) 3.00 ms^{-2} (c) 4.00 ms^{-2} (d) 9.00 ms^{-2}
6. A moving object is said to have uniform acceleration if its (a) displacement decreases at a constant rate (b) speed is directly proportional to time (c) velocity increases by equal amount in equal time intervals (d) velocity varies inversely with time
7. The diagram shows a velocity-time graph of the motion of a car. What is the total distance covered after the journey? (a) 75m (b) 150m (c) 300m (d) 375m
8. The area under a velocity-time graph represents (a) final velocity attained (b) direct covered (c) acceleration (d) workdone
9. A body accelerates uniformly from rest at 2 ms^{-2} . Calculate its velocity after travelling 9m. (a) 36 ms^{-1} (b) 18 ms^{-1} (c) 6 ms^{-1} (d) 4.5 ms^{-1}
10. A moving object is said to have uniform acceleration if its
(a) displacement decreases at a constant rate (b) speed is directly proportional to time
(c) velocity increases by equal amount in equal time intervals (d) velocity varies inversely with time

THEORY

1. A body moving with uniform acceleration a , has two points (5, 15) and (20, 60) on the velocity-time graph of its motion. Calculate the acceleration a .
2. Two points on a velocity-time graph coordinates (5s, 10 ms^{-1}) and (20s, 20 ms^{-1}). Calculate the mean acceleration between the two points.
3. A car starts from rest and accelerates uniformly for 5s until it attains a velocity of 30 ms^{-1} . It then travels with uniform velocity for 15s before decelerating uniformly to rest in 10s;
(i) Sketch a graph of the motion
(ii) Using the graph above, calculate the
(a) Acceleration during the first 5s
(b) Deceleration during the last 10s
(c) Total distance covered through the motion
4. A car starts from rest and accelerates uniformly for 10s, until it attains a velocity of 25 m/s , it then travels with uniform velocity for 20s before decelerating uniformly to rest in 5s.
(i) Calculate the deceleration during the last 5s
(ii) Calculate the acceleration during the first 10s
(iii) Sketch a graph of the motion and calculate the total distance covered throughout the motion.
5. (a) Using a suitable diagram, explain how the following can be obtained from a velocity-time graph
(i) Acceleration (ii) Retardation (iii) Total distance
(b) Show that the displacement of a body moving with uniform acceleration a is given by $S = ut + \frac{1}{2}at^2$, where u is the velocity of the body at time $t=0$

- (c) A particle moving in a straight line with uniform deceleration has a velocity of 40m/s at a point P, 20m/s at a point Q and comes to rest at a point R, where QR=50m. Calculate the:
- (i) Distance PQ (ii) Time taken to cover PQ (iii) Time taken to cover PR (WAEC, 1990)
6. (a) What is meant by the statement *the acceleration of free fall due to gravity on the equator is 9.78ms^{-2}*
- (b) State two factors that affect the value of the acceleration due to gravity.(WAEC,2006)
7. Using suitable diagram, explain how the following can be obtained from a velocity- time graph: (a)Acceleration (b) Total distance covered (c) A body at rest is given an initial uniform acceleration of 6.0ms^{-2} for 20s after which the acceleration is reduced to 4.0ms^{-2} for the next 10s. The body maintains the speed attained for 30s. Draw the velocity-time graph of the motion using the information provided above. From the graph, calculate the:
- (i) Maximum speed attained during the motion
- (ii) Total distance travelled during the first 30s
- (iii) Average speed during the same time interval as in (ii) above (WAEC, 2009)
8. (a) Sketch a distance-time graph for a particle moving in a straight line:
- (i) Uniform speed (ii) Variable speed (NECO, 1010)
- (b) A body starts from rest and travels distances of 120, 300, and 800m in successive equal time intervals of 12s. During each interval the body is uniformly accelerated.
- (i) Calculate the velocity of the body at the end of each successive interval.
- (ii) Sketch the velocity- time graph of the motion. (WAEC, 2010)
9. (a) Explain the terms: uniform acceleration and average speed.
- (b) A body at rest is given an initial uniform acceleration of 8.0ms^{-2} for 30s after which the acceleration is reduced to 5.0ms^{-1} for the next 20s. The body maintained the speed attained for 60s after which it is brought to rest in 20s.Draw the velocity-time graph of the motion using the information given above.
- (c) Using the graph, calculate the:
- (i) Maximum speed during the motion.
- (ii) Average retardation as the body is being brought to rest.
- (iii) Total distance travelled during the first 50s.
- (iv) Average speed during the same interval as in (ii) above (WAEC, 1991)
10. (a) State two reasons why the acceleration due to gravity varies on the surface of the earth. (NECO, 2008)
- (b) State the difference between centripetal and centrifugal force.(NECO, 2011)
11. (i) Define velocity and acceleration
- (ii)List two physical quantities that can be deduced from a velocity-time graph.
- Define the following terms; (a) average speed (b) Instantaneous velocity
- (c) A car travels at an average speed of 20ms^{-1} . Calculate the distance covered in 1hour
- 12 Starting from rest, a vehicle accelerates at 2m/s^2 for 5secs it then travels for 5secs at the velocity, V_0 reached and is brought to rest with a uniform retardation after the next 5s.
- a. Sketch the velocity-time graph for the journey
- b. Calculate the value of V_0 ,
- c. What is the retardation
- d. The total distance covered

STUDY SESSION SIX: DENSITY & RELATIVE DENSITY

INTRODUCTION

This session introduces the concepts of **density** and **relative density**, two important physical properties used to describe and compare different substances. It explains density as the relationship between the mass of a substance and the volume it occupies, making it possible to determine how compact or closely packed the particles of a material are. The session also discusses **relative density**, which compares the density of a substance with that of water and is expressed without a unit.

In addition, the session examines the methods used to determine the density of solids and liquids, explains the factors that influence density, and highlights the practical applications of density and relative density in science, engineering, medicine, and everyday life. A proper understanding of these concepts provides a strong foundation for the study of buoyancy, flotation, and fluid mechanics.

LEARNING OUTCOMES FOR THE STUDY SESSION

When you have studied this session, you should be able to:

- i. Define density and relative density.
- ii. Distinguish between density and relative density.
- iii. Calculate the density of solids and liquids.
- iv. Determine the relative density of substances.
- v. State the SI unit of density.
- vi. Identify factors that affect the density of materials.
- vii. Explain the practical applications of density and relative density in everyday life and science.

6.1 DEFINITION OF DENSITY

The density of a substance is the mass per unit volume of the substance.

$$\text{Density} = \frac{\text{mass of a given substance}}{\text{Volume of the substance}}$$

Density is scalar quantity & measured in kgm^{-3} (kilogram per cubic meter)

Determination of Density

The determination of density involves the determination of a mass and a volume. The mass can be found by weighing. The density of a substance can be determined using a graduated density bottle.

Relative Density

Relative density is also known as specific gravity. Relative density of a substance is defined as the density of the substance per density of water.

$$\text{R.D} = \frac{\text{Density of the substance}}{\text{Density of water}}$$

R.D is also equal to the ratio weight of a substance to weight of an equal volume of water. As weight is proportional to mass

$$\text{R.D} = \frac{\text{mass of substance}}{\text{mass of equal volume of water}}$$

Determination of R.D of Solid (e.g. Sand)

Mass of empty bottle = m_1

Mass of bottle + sand = m_2

Mass of bottle + sand + water = m_3

Mass of bottle + water only = m_4

Mass of sand = $m_2 - m_1$

Mass of water added to sand = $m_3 - m_2$

Mass of water filling the bottle = $m_4 - m_1$

Mass of water having the same volume as sand = $(M_4 - M_1) - (M_3 - M_2)$

Relative density =
$$\frac{\text{Mass of sand}}{\text{Mass of equal volume of water}}$$

$$\text{R.D} = \frac{m_2 - m_1}{(m_4 - m_1) - (m_3 - m_2)}$$

EVALUATION

1. Differentiate between density & relative density
2. A glass block of length 100cm width 60cm and thickness 20cm has a mass of 4000g. calculate the density of the glass

6.2 Determination of R.D of Liquid

mass of empty density bottle = m_1

mass of bottle filled with water = m_2

mass of bottle filled with liquid = m_3

$$\text{R.D of liquid} = \frac{m_3 - m_1}{m_2 - m_1}$$

Example - A glass block of length 10cm width 8cm and thickness 2cm has a mass of 400g. calculate the density of the glass.

Solution

$$l = 10\text{cm} = 0.1\text{m}, b = 8\text{cm} = 0.08\text{m}, h = 2\text{cm} = 0.02\text{m}, m = 400\text{g} = 0.4\text{kg}$$

$$V = lbh = 0.1 \times 0.08 \times 0.02 = 0.00016\text{m}^3$$

$$\text{Density} = \frac{\text{Mass (m)}}{\text{Volume (V)}} = \frac{0.4}{0.00016} = 2500\text{kgm}^{-3}$$

Example - Calculate the volume in m^3 of a piece of wood of mass 500g and density 0.76 gcm^{-3}

mass of the wood = 500g

density = 0.76gcm^{-3}

volume = ?

$$\text{volume} = \text{mass} / \text{density} = \frac{500}{0.76}$$

$$= 658\text{cm}^3$$

$$\text{volume} = 658\text{cm}^3 = 6.58 \times 10^{-4} \text{ m}^3$$

Example - An empty relative density bottle has a mass of 15.0g. when completely filled with water, its mass is 39.0g. what will be its mass if completely filled with acid of relative density 1.20?

solution

m_1 , mass of empty bottle = 15.0g

m_2 , mass of bottle + water = 39.0g

mass of acid = $n - 15.0\text{g}$

$$\begin{aligned} \text{mass of water} &= 39.0 - 15.0\text{g} \\ &= 24.0\text{g} \end{aligned}$$

$$\text{R.D} = 1.20$$

$$\text{R.D} = \frac{n - 15.0\text{g}}{39.0 - 15.0\text{g}}$$

$$1.20 = \frac{n - 15.0}{24.0}$$

$$n - 15.0 = 1.20 \times 24.0$$

$$n - 15 = 28.8$$

$$n = 28.8 + 15$$

$$n = 43.8\text{g}$$

NB: The hydrometer is an instrument used to measure the relative density of liquids

EVALUATION

1. The volume of an object is $1.5 \times 10^3 \text{ m}^3$ and its mass is $3.0 \times 10^4 \text{ kg}$. Calculate its density.
2. A relative density bottle weighs 20g when empty, 80g when filled with water & 100g when filled with liquid. Find the relative density of the liquid.

READING ASSIGNMENT

New school physics by M.W.Anyakoha,Phd.Pg 152 – 157

WEEKEND ASSIGNMENT

1. Find the density of a substance, if the mass of the substance is 150,000g and the dimension is 20m by 10m by 500cm.
a. 0.5 kg/m^3 b. 0.24 kg/m^3 c. 1.50 kg/m^3 .
2. What is the height of a cylindrical iron if the density is 7900 kg/m^3 ? The mass is 700kg and the radius is 0.1m.
a. 2.918m b. 2.819m c. 3.418m.
3. Density is defined as the ratio of mass to-----
a. Pressure b. area c. volume
4. Relative density is the ratio of mass of a substance to-----
a. Mass of 2an equal volume of water b. volume of a substance c. density
5. The S.I unit of density is ----- a. g/cm^3 b. kg/m^3 c. kg/m .

THEORY

1. Alcohol of mass 33.2g and density 790 kg/m^3 is mixed with 9g of water. What is the density of the resulting mixture?(density of water is 1 g/cm^3).
2. Define relative density of liquid.

STUDY SESSION SEVEN: PRESSURE , ARCHIMEDES' PRINCIPLES, UPTHURST & LAWS OF FLOATATION

INTRODUCTION

This session introduces some of the fundamental concepts of fluids in Physics, with emphasis on **pressure, upthrust, Archimedes' Principle, and the law of floatation**. These concepts explain how liquids and gases exert forces on objects and why some objects float while others sink. The study of fluid behavior has numerous practical applications in engineering, transportation, medicine, marine technology, and everyday life. By understanding these principles, students will be able to explain common phenomena such as the floating of ships, the operation of hydraulic systems, the use of life jackets, and the behavior of submerged objects in different fluids. This session also develops learners' problem-solving skills through the application of these principles to real-life situations and numerical calculations.

LEARNING OUTCOMES FOR THE STUDY SESSION

When you have studied this session, you should be able to:

- i. Define pressure and state its SI unit.
- ii. Explain the factors affecting pressure in solids and fluids.
- iii. Describe the concept of upthrust and identify the factors that influence it.
- iv. State and explain Archimedes' Principle.
- v. Apply Archimedes' Principle to solve simple numerical problems.
- vi. Explain the law of floatation and distinguish between floating and sinking objects.
- vii. Relate pressure, upthrust, and floatation principles to practical applications in everyday life, engineering, and marine transportation.

7.0 PRESSURE

Pressure is defined as the perpendicular force per unit area acting on a surface. It is a scalar quantity & measured in N/m^2 or Pascal (pa). It can also be defined as the force per unit area, which is calculated by taking the total force and dividing it by the area over which the force acts. Force and pressure are related but different concepts. A very small pressure, if applied to a large area, can produce a large total force.

$$P = \frac{F}{A} \dots\dots\dots 1. \quad \text{Where } P\text{-pressure, } F\text{- force (N) \& } A\text{-area (m}^2\text{)}$$

NB: $1 \text{ bar} = 10^5 \text{ N/m}^2 = 10^5 \text{ pa}$

Example – A force of 40N acts on an area of 5m^2 . What is the pressure exerted on the surface?

Solution

$$F = 40\text{N}, A = 5\text{m}^2, P = ?$$

$$P = F/A = 40/5 = 8\text{pa}$$

7.1 Pressure in Liquid

Pressure in liquid has the following properties

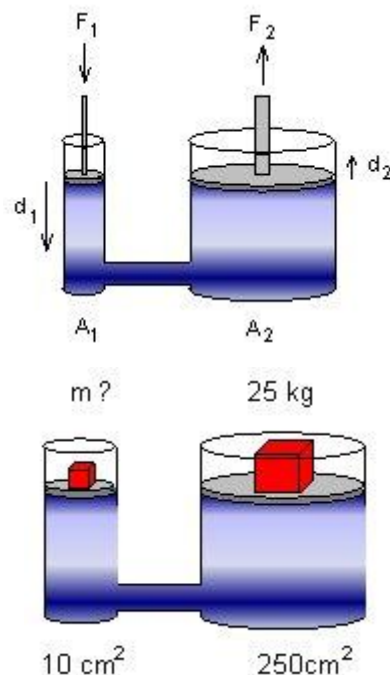
1. Pressure increases with depth
2. Pressure depend on density
3. Pressure at any point in the liquid acts equally in all direction
4. Pressure at all points at the same level within a liquid is the same
5. It is independent of cross-sectional area

$$P = h\rho g \dots\dots\dots 2.$$

where p-pressure, h-height & g-acceleration due to gravity

Pascal's principle : Pressure applied to an enclosed fluid is transmitted undiminished to every part of the fluid, as well as to the walls of the container. The operation of the hydraulic press & the car brakes system is based on this principle.

The ideal press consists of two pistons of areas (a , A) enclosed between them incompressible liquid as in figure



When a small force (f) acts on the small piston (a) , it exerts a pressure (p = f/a).

To keep the large piston (A) at equilibrium with the small one (a) a load = F is placed on the large piston .

$$P = f/a = F/A$$

EVALUATION

1. Define pressure
2. State five characteristics of pressure in liquid

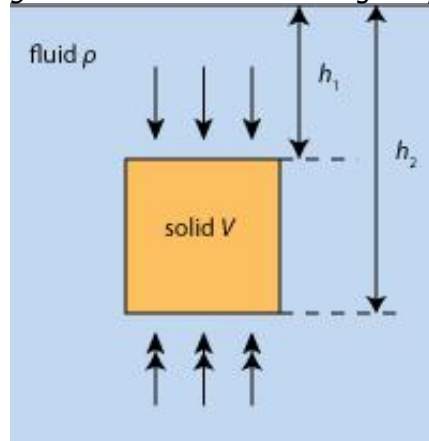
7.2 Archimedes' Principle & Upthrust

Archimedes' principle is a law that explains buoyancy or upthrust. It states that When a body is completely or partially immersed in a fluid it experiences an upthrust, or an apparent loss in weight, which is equal to the weight of fluid displaced. According to a tale, Archimedes discovered this law while taking a bath. An object experiences upthrust due to the fact that the pressure exerted by a fluid on the lower surface of a body being greater than that on the top surface, since pressure increases with depth. Pressure, p is given by $p = h\rho g$, where:

h is the height of the fluid column

ρ (rho) is the density of the fluid

g is the acceleration due to gravity



Let us confirm this principle theoretically. On the figure on the left, a solid block is immersed completely in a fluid with density ρ . The difference in the force exerted, d on the top and bottom surfaces with area a is due to the difference in pressure, given by

$$d = h_2 a \rho g - h_1 a \rho g = (h_2 - h_1) a \rho g$$

But $(h_2 - h_1)$ is the height of the wooden block. So, $(h_2 - h_1)a$ is the volume of the solid block, V .

$$d = V \rho g$$

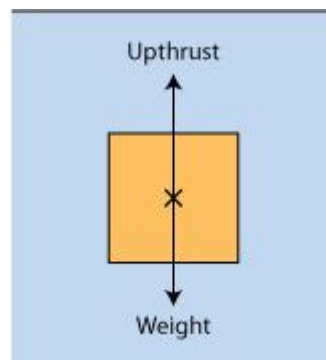
$$\text{Upthrust} = V \rho g$$

In any situation, the volume of fluid displaced (or the volume of the object submerged) is considered to calculate upthrust, because $(h_2 - h_1)$ is the height of the solid block only when it is completely immersed. Furthermore, the pressure difference of the fluid acts only on the immersed part of an object.

Now, moving back to $V\rho g$. Since V is the volume of fluid displaced, then the product of V , ρ and g is the weight of the fluid displaced. So, we can say that

$$\text{Upthrust} = \text{Weight of the fluid displaced}$$

Compare this conclusion with the statement above summarising Archimedes' principle. Are they the same? Well, not totally. The "apparent loss in weight" was not mentioned.



In the figure on the left, there are arrows on the top and bottom of the solid block. The downward arrow represent the weight of the block pulling it downwards and the upward arrow represent the upthrust pushing it upwards. If one were to measure the weight of the solid block when it is immersed in the fluid, he will find that the weight of the block is less than that in air. There is a so-called "apparent loss in weight", because the buoyant force has supported some of the block's weight.

NB: 1. When an object is wholly immersed, it displaces its volume of fluid. So up thrust = weight of fluid displaced. = Volume of fluid displaced x its density x g = volume of object x density of fluid x g

2 When the object is partially immersed e.g. if $\frac{1}{4}$ of its volume (v) is immersed then the up thrust is given by $\frac{v}{4} \times \text{density of liquid} \times g$.

7.3 Determination of Relative Density by Archimedes' Principle

1. Relative density of solid

The body is weighed in air w_1 , and then when completely immersed in water w_2

$$\begin{aligned} \text{Relative density of solid} \\ &= \frac{\text{Weight of solid in air}}{\text{Weight of equal volume in water}} \\ &= \frac{w_1}{w_1 - w_2} \end{aligned}$$

2. Relative density of liquid

A solid is weighed in air (w_1), then in water (w_2) and finally in the given liquid (w_3)

$$\text{Relative density of liquid} = \frac{\text{apparent loss of weight of solid in liquid}}{\text{apparent loss of weight of solid in water.}}$$

$$= \frac{w_1 - w_3}{w_1 - w_2}$$

Example - The mass of a stone is 15g when completely immersed in water and 10g when completely immersed in liquid of relative density 2.0 . What is the mass of the stone in air?

Solution:

$$\text{Relative density} = \frac{\text{upthrust in liquid}}{\text{upthrust in water}}$$

let W represents the mass of the stone in air

$$2 = \frac{w - 10}{w - 15}$$

$$2(w - 15) = w - 10$$

$$2w - 30 = w - 10$$

$$2w - w = -10 + 30$$

$$w = 20g$$

7.4 Law of Floatation

A floating object displaces its own weight of the fluid in which it floats or an object floats when the upthrust exerted upon it by the fluid is equal to the weight of the body. When an object is floating freely (i.e. neither sinking nor moving vertically upwards), then the upthrust must be fully supporting the object's weight. We can say

Upthrust on body = Weight of floating body. By Archimedes' principle,

Upthrust on body = Weight of fluid displaced. Therefore, Weight of floating body = Weight of fluid displaced

This result, sometimes called the "principle of floatation", is a special case of Archimedes' principle

EVALUATION

1. State the law of floatation.
2. State Archimedes' principle.

READING ASSIGNMENT

www.google.com (click on google search, type "Archimedes' principle", click on search) & New sch. physics by M.W.Anyakoha,Phd. Pg 348 – 358, 150 - 152

WEEKEND ASSIGNMENT

1. A force of 40N acts on an area of 10m^2 . What is the pressure exerted on the surface?
(a) 8pa (b) 4pa (c) 400pa (d) 10pa

2. What is the height of a cylindrical iron if the density is 7900kg/m^3 the mass is 700kg and the radius is 0.1m [a] 2.918cm [b] 2.819m © 3.418m
3. Density is defined as the ratio of mass to (a) pressure (b) area (c) volume
4. Relative density is the ratio of mass of a substance to ----- (A) mass of equal volume of water (b) volume of a substance (c) density
5. Pressure can be measured in the following except (a) bar (b) N/m^2 (c) pascal (d) Nm^2

THEORY

1. Differentiate between force & pressure.
2. What is the pressure due to water at the bottom of a tank which is 20cm deep and is half of water? (Density of water = 10^3kg/m^3 and $g = 10\text{m/s}^2$)

STUDY SESSION EIGHT: WORK, ENERGY AND POWER

INTRODUCTION

Work, energy, and power are among the most important concepts in Physics because they explain how forces produce motion, how energy is transferred from one form to another, and how quickly work is performed. These concepts are encountered in everyday activities such as walking, lifting objects, operating machines, driving vehicles, and generating electricity. Understanding the relationship between work, energy, and power provides a solid foundation for explaining many natural phenomena and technological applications.

In this study session, students will learn the meaning of work, energy, and power, the conditions necessary for work to be done, and the various forms of energy found in nature. The session also examines work done in a force field, the principle of conservation of energy, and the processes through which energy is transformed from one form to another. By studying these concepts, learners will gain a deeper understanding of how energy is utilized efficiently in everyday life, engineering systems, and industrial applications.

LEARNING OUTCOMES FOR THE STUDY SESSION

When you have studied this session, you should be able to:

- i. Define work, energy, and power correctly.
- ii. Explain conditions necessary for work done.
- iii. Calculate work, energy, and power accurately.
- iv. Identify different forms of energy.
- v. Explain work done in force fields.
- vi. Describe energy conversion in practical situations.
- vii. State and apply the law of conservation of energy.
- viii. Solve numerical problems involving work, energy, and power.

8.1 WORK

Work is said to be done whenever a force moves a body through a certain distance in the direction of force. Work done can be defined mathematically as the product of the force and the displacement. It is a scalar quantity & measured in Joules

$$W = F \times d \text{1.}$$

$$W = mgh \text{2.}$$

If a force is applied on a body at an angle θ to the horizontal

$$\text{Work done to move the body on the horizontal plane} = F \cos \theta \times d \text{3.}$$

$$\text{Work done to raise the body to an appreciable height} = F \sin \theta \times d \text{4.}$$

Examples- A boy of mass 50kg runs up a set of steps of total height 3.0m. Find the work done against gravity

Solution

$$m = 50\text{kg}, h = 3\text{m}, g = 10\text{m/s}^2$$

$$\text{Work done} = mgh$$

$$= 50 \times 10 \times 3$$

$$= 1500 \text{ Joules}$$

8.2 Energy

Energy is defined as the ability to do work. It is a scalar quantity & measured in Joules. There are many forms of energy. These include:

- i. Mechanical energy
- ii. Thermal energy
- iii. Chemical energy
- iv. Electrical energy
- v. Nuclear/Atomic Energy
- vi. Solar/Light energy
- vii. Sound Energy

8.3 Types of Mechanical Energy

Mechanical energy is classified as

- 1) Potential energy
- 2) Kinetic energy

8.4 POTENTIAL ENERGY:- is simply "stored energy" i.e. energy possessed by a body by virtue of its states:

$$P.E = mgh \dots\dots\dots 5.$$

8.5 KINETIC ENERGY: is the energy possessed by a body by virtue of its motion.

Examples are a student running a race, wind or air motion, electrical charges in motion, a moving bullet

$$K.E = \frac{1}{2} mv^2 \dots\dots\dots 6.$$

Example - An object of mass 5kg is moving at a constant velocity of 15m/s. Calculate its kinetic energy.

Solution:

$$K.E = \frac{1}{2} mv^2 = \frac{1}{2} \times 5 \times 15 \times 15 = 562.5 \text{ J}$$

Example - Find the potential energy of a boy of mass 10kg standing on a building floor 10m above the ground level. $g = 10\text{m/s}^2$

Solution:

$$P.E = mgh = 10 \times 10 \times 10 = 1000 \text{ J}$$

8.6 POWER

Power is defined as the rate of doing work or the rate of transfer of energy. It is a scalar quantity & measured in watt

$$\text{Power} = \frac{\text{work done}}{\text{Time}} \dots\dots\dots 7$$

$$P = (F \times d)/t = F \times d/t = FV \dots\dots\dots 8$$

Example: - : A boy of mass 10kg climbs up 10 steps each of height 0.2m in 20 seconds. Calculate the power of the boy.

Solution

$$\text{Height climbed} = 10 \times 0.2 = 2\text{m}$$

$$\text{Work done} = mgh = 10 \times 10 \times 2 = 200 \text{ Joules}$$

$$\text{POWER} = \frac{\text{work}}{\text{Time}} = \frac{10 \times 10 \times 2}{20} = 10\text{watts}$$

EVALUATION

1. Define power.
2. A boy of mass 960g climbs up to 12 steps each of height 20cm in 20 seconds. Calculate the power of the boy.

8.7 WORK DONE IN A FORCE FIELD & ENERGY CONVERSION

Work done in Lifting a Body & Falling Bodies

The magnitude of work done in lifting a body is given by

$$\text{Work} = \text{force} \times \text{distance} = mg \times h = mgh$$

Also, the work done on falling bodies is given by

$$\text{Work} = \text{force} \times \text{distance} = mg \times h = mgh$$

EVALUATION

1. Explain three types of force field.
2. A loaded sack of total mass 100kg falls down from the floor of a lorry 2m high. Calculate the work done by gravity on the load.

8.8 Conservation & Transformation of Energy

Energy can be converted from one form to another in a closed system. The law of conservation of energy states that in an enclosed system, energy can neither be created nor destroyed during transformation. Examples of such conversions include (a) Motor converts electrical to mechanical

energy (b) Generator converts mechanical to electrical energy (c) Electric pressing iron convert electrical to heat energy

8.9 World Energy Resources

World energy resources can be classified as

1. Renewable Energy Resources: They are energy that can be replaced as they are used e.g. solar energy, wind energy, water energy & biomass
2. Non-renewable Energy Resources: Energy that cannot be replaced after use e.g. nuclear energy, petroleum & natural gas

EVALUATION

1. State the law of conservation of energy.
2. Differentiate between renewable & non-renewable energy.

READING ASSIGNMENT

New school physics by M.W.Anyakoha,Phd.Pg 29, 30 & 34

GENERAL EVALUATION

1. When is work said to be done?
2. State the difference between work, energy and power.
3. Differentiate between kinetic and potential energy.
4. State the energy transformation that take place during electricity generation at kanji dam.

WEEKEND ASSIGNMENT

1. The following are example of force field except (a) electric force (b) magnetic force (c) frictional force (d) gravitational force
2. Electric cell convert to electrical energy (a) nuclear (b) chemical (c) mechanical (d) heat
3. The following are examples of renewable energy except (a) biomass (b) solar (c) wind (d) nuclear
4. A boy of mass 50kg runs up a set of steps of total height 3.0m. Find the work done against gravity (a) 1200J (b) 1500J (c) 1000J (d) 1300J
5. The SI unit of power is (a) joules (b) kilogram (c) watt (d) pascal
- 6 A bob of a simple pendulum has a mass of 0.02kg. Determine the weight of the bob (a) 0.2w (b) 0.52w (c) 0.25w (d) 2N
- 7 An object of mass 0.5kg has K.E of 25J. calculate the speed of the object (a) 50ms^{-1} (b) 25ms^{-1} (c) $2.\text{ms}^{-1}$ (d) 10ml^{-1}
- 8 An object of mass 0.5kg has a velocity of 4ms^{-1} Calculate the K.E (a) 4.0J (b) 40J (c) 0.4J (d) 400J
- 9 Which of the following fundamental quantities is not correctly paired with its unit of measurement? (a) Electricity current – Ampere (b) Amount of substance – kilogram (c) Temperature – Kelvin (c) length – meter
- 10 A diver is 5.2m below the surface of water of density 10^3 kg/m^3 . If the atmospheric pressure is $1.02 \times 10^5 \text{ pa}$. Calculate the pressure on the diver. [$g=10\text{mls}^2$) (a) $6.02 \times 10^4 \text{ pa}$ (b) $1.02 \times 10^5 \text{ pa}$ (c) $1.54 \times 10^5 \text{ pa}$ (d) $5.20 \times 10^5 \text{ pa}$

THEORY

- 1 Explain work done.
- 2 A boy of mass 960g climbs up to 12 steps each of height 20cm in 20 seconds. Calculate the power of the boy.
- 3 A loaded sack of total mass 100kg falls down from the floor of a lorry 2m high. Calculate the work done by gravity on the load.
- 4 State the law of conservation of energy.

STUDY SESSION NINE: VISCOSITY

INTRODUCTION

Viscosity is an important property of fluids that describes their resistance to flow. It explains why some liquids, such as honey and engine oil, flow more slowly than water, while gases also exhibit varying degrees of internal resistance during motion. The concept of viscosity is fundamental in Physics because it helps explain the behavior of fluids in both natural and industrial processes.

In this study session, students will learn the meaning of viscosity, the factors that affect it, and the differences between the flow of ideal and real fluids. The session also examines the effects of temperature on viscosity, the concept of terminal velocity, and the practical applications of viscosity in engineering, medicine, transportation, and everyday life. A proper understanding of viscosity enables learners to explain fluid motion and appreciate its importance in scientific and technological developments.

LEARNING OUTCOMES FOR THE STUDY SESSION

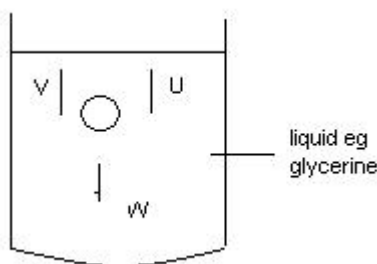
When you have studied this session, you should be able to:

- Define viscosity and explain its meaning.
- Differentiate between viscous and non-viscous fluids.
- Identify factors affecting the viscosity of fluids.
- Explain the effect of temperature on viscosity.
- Describe the concept of terminal velocity.
- State the practical applications of viscosity.
- Solve simple problems involving viscosity.
- Relate viscosity to everyday and industrial applications.

9.1 MEANING OF VISCOSITY

Viscosity is the internal friction which exists between layers of the molecules of a fluid (liquid or gas) in motion. The viscosity of a fluid can also be defined as the measure of how resistive the fluid is to flow. It is a vector quantity & measured in pascal-seconds(pa.s). It can be defined mathematically as the ratio of the **shearing stress** to the **velocity gradient** in a fluid

$$\text{Viscosity } (\eta) = \frac{\text{Force}}{\text{Area} \times \text{Velocity gradient}} \dots\dots\dots 1.$$
$$\text{Velocity gradient} = \frac{\text{velocity}}{\text{Length}} \dots\dots\dots 2$$



$$W = U + V$$
$$W - U - V = 0 \dots\dots\dots 3.$$

$V = W - U$ (apparent or effective weight) where V-viscous force, W- weight, U- upthrust

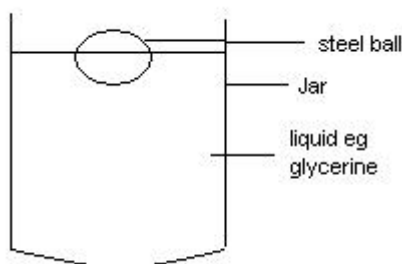
NB : Substances with low viscosity include water, kerosene, petrol, ethanol. Those with high viscosity are glue, syrup, grease, glycerine etc

9.2 Experiment to Determine the Terminal Velocity of a Steel Ball Falling Through a Fluid

Aim: To determine the terminal velocity of a steel ball falling in through a jar of glycerin

Apparatus: steel ball, cylindrical calibrated jar, glycerine

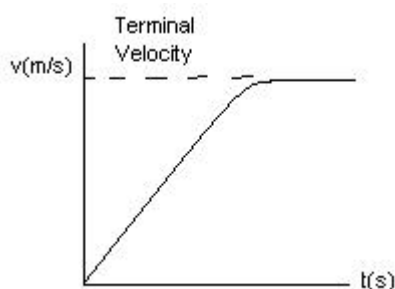
Diagram:



Procedure: Set-up the apparatus as shown above & gently drop the steel ball in the jar of glycerin

Observation: It will be observed that the ball is accelerating in the liquid. Also the time taken for the ball to move from A-B will be different from B-C and so on. A time will be reached when the ball will be moving at a constant speed or velocity. It is that point that terminal velocity is experienced.

Graph :



Conclusion: Terminal velocity is attained when $W = V + U$. At a point when the ball is moving at a constant speed through the glycerine.

Precaution: 1. The steel ball should be dropped gently on the liquid 2. Experiment should be done under constant temperature 3. Avoid error of measurement when taken the reading.

NB: **Terminal velocity** is the maximum velocity of an object when the viscous force due to motion of the object equals the apparent (effective) weight of the object in the fluid where there is no longer net force on the object.

Drag force is the force that keeps the object continuously moving after the terminal velocity has been attained.

Stokes' Law states that at the terminal velocity, the upward frictional force (F) = $6\pi \eta rV$
Where F - Frictional/ Drag force, η - viscosity, r - radius of sphere, V_t - Terminal velocity

EVALUATION

1. Derive the dimension of viscosity.
2. Describe an experiment to determine the terminal velocity of a steel ball falling in a fluid.

Factors Affecting Viscosity

1. Viscosity varies with material (viscosity is a property of material)
2. The viscosity of simple liquids (a) decreases with increasing temperature (b) increases under very high pressure
3. The viscosity of gases (a) increases with increasing temperature (b) is independent of pressure & density

Effect of Viscosity

1. Viscosity is responsible for different rate of fluid flow.
2. Viscosity affects motion of body in fluid.

9.3 Application of Viscosity

1. It is used as a lubricant.
2. The knowledge of viscous drag/drag force is applied in the design of ship & aircraft.
3. Used to estimate the enlarged size of particles.

EVALUATION

1. State two effects of viscosity.
2. State two applications of viscosity.

GENERAL EVALUATION

1. What is viscosity?
2. What is terminal velocity?
3. State two(2) substances with high viscosity.
4. State Stoke's law.
5. What is a viscostatic substance?

READING ASSIGNMENT

New school physics by M.W.Anyakoha,Phd.Pg 105 – 107

WEEKEND ASSIGNMENT

1. Viscosity opposes motion of an object in (a) solid (b) liquid only(c) gas only (d) liquid & gas
2. The SI unit of velocity gradient is (a) m/s (b) s^{-1} (c) m/s^2 (d) ms
3. Terminal velocity is attained when (a) $w + v = u$ (b) $w = v - u$ (c) $w + u = v$ (d) $w = v + u$
4. The following are vector quantities except (a) friction (b) viscosity (c) upthrust (d) pressure
5. Friction and viscosity are similar but not the same. True/False

THEORY

1. Explain viscosity.
 2. Describe an experiment to determine the terminal velocity of a steel ball falling in a fluid.
- NB: A liquid is said to be **VISCOSTATIC** if its viscosity does not change (appreciably) with change in temperature.

SUMMARY

This course has provided a comprehensive introduction to the fundamental principles of Physics and their applications in everyday life, science, and technology. It began with the study of units, dimensions, and measurements, which form the foundation of all scientific investigations. Students were introduced to the International System of Units (SI), the distinction between fundamental and derived quantities, and the importance of accurate measurement of physical quantities such as mass, weight, length, and time.

The course also explored the concept of motion in nature, including force, circular motion, and the effects of centripetal and centrifugal forces. It examined friction as a force that opposes motion and discussed its advantages, disadvantages, and practical applications. In addition, learners studied vector and scalar quantities, distance and displacement, speed and velocity, acceleration, motion graphs, and the equations of uniformly accelerated motion, enabling them to analyze and solve problems involving moving objects.

Furthermore, the course covered the properties of matter by examining density and relative density, as well as the behavior of fluids through pressure, Archimedes' principle, upthrust, and the laws of floatation. Students also gained an understanding of the concepts of work, energy, and power, including energy transformation, conservation of energy, and work done in force fields. Finally, the study of viscosity introduced learners to the resistance of fluids to flow, the factors affecting viscosity, and its significance in science, engineering, medicine, and various industrial applications.

Overall, this course has established a strong foundation in Physics by combining theoretical knowledge with practical applications. The concepts studied serve as essential building blocks for advanced studies in Physics, Engineering, Medicine, Environmental Science, and other science-related disciplines. Students are encouraged to continue practicing numerical problems, conducting laboratory experiments, and applying the principles learned to real-life situations in order to strengthen their understanding and develop sound scientific reasoning.